Certifiable Trajectory Optimization for Locomotion via Semidefinite Relaxation

Scientific thesis for the procurance of the degree M.Sc.
(Intermediate report)
from the TUM School of Computation, Information and Technology
at the Technical University of Munich.

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Submitted on Munich, 13.05.2026

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LEHRSTUHL FÜR INFORMATIONSTECHNISCHE REGELUNG
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External Institution: Harvard SEAS — Computational Robotics Group

January, 12 2026

MASTER'S THESIS

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Problem description:

Nonlinear discretized dynamics and nonconvex geometric constraints yield a highly nonconvex optimization landscape; in contact-rich regimes, the discontinuity of contact forces further degrades it [4]. Classical approaches typically rely either on nonlinear programming (which is highly sensitive to initialization) [5] or on mixed-integer formulations [1] (which incur worst-case exponential complexity).

Locomotion requires reasoning about nonlinear dynamics, discontinuous contacts, and nonconvex geometry (self/obstacle avoidance). These properties create hard optimization landscapes and brittleness to initialization. In contrast, Moment-SOS semidefinite relaxation offers: (1) a convex lower bound that certifies suboptimality; (2) principled extraction of candidate trajectories; and (3) natural ways to exploit problem sparsity (chain-like temporal structure and robot-specific sparsity from kinematic chains and separable contacts) [2, 6]. Specialized first-order SDP solvers (including GPU-based ADMM variants) make these relaxations practically viable [3]. This thesis translates these ideas into legged locomotion on real hardware, using a certifiable trajectory optimization framework for locomotion tasks via sparse semidefinite relaxation.

Tasks:

- Review relevant literature on Lie-group based dynamics modeling for legged locomotion, including contact modeling and polynomial optimization methods on manifolds.
- Derive and formulate discrete-time Lie-group variational dynamics and express objectives and constraints as polynomial polynomialized optimization problems exploiting system structure.
- Develop sparse Moment-SOS / semidefinite relaxations that leverage temporal, kinematic, and contact sparsity for efficient and certifiable trajectory optimization.
- Evaluate certifiable solvers with trajectory extraction and validate the approach in simulation with respect to performance, robustness, runtime, and certification gap.

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Start: 01.03.2026
Intermediate Report: 13.05.2026
Delivery: 01.09.2026

Abstract

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2.1 Special Matrix Lie Groups

2.2 Special Lie Algebras

Proposition 1 (Differential of left translation on a matrix Lie group: $T_x L_h$)**.** *Let $G \subset GL(n, \mathbb{R})$ be a matrix Lie group. For $h \in G$, define $L_h : G \to G$ by $L_h(g) = hg$ [Hal00]. Then for each $x \in G$ the differential*

$$T_x L_h : T_x G \to T_{hx} G,$$

of the left translation map $L_h : G \to G$ is given by

$$T_x L_h(v) = hv. \tag{2.1}$$

Proof. By the definition of the tangent space of an embedded submanifold given in [Hal00], there exists a smooth curve $\gamma : (-\epsilon, \epsilon) \to G$ with $\gamma(0) = x$ and the tangent vector $\dot{\gamma}(0) = v \in T_x G$. In [**?**], the differential is defined as

$$T_x L_h(v) = \frac{d}{dt} L_h(\gamma(t))\bigg|_{t=0} = \frac{d}{dt}(h\gamma(t))\bigg|_{t=0},$$

with the left translation map $L_h(g) = hg$. Since h is constant and $\dot{\gamma}(0) = v$, it follows that

$$T_x L_h(v) = h\dot{\gamma}(0) = hv.$$

□

It can be observed that $L_h(g) = hg$ and $T_x L_h(v) = hv$ have the same matrix expression, but acting on different objects. The left translation map $L_h(g)$ is multiplying group elements $g \in G$, while the differential of the left translation map is the pushforward of tangent vectors $v \in T_x G \subset \mathbb{R}^{n \times n}$. The differential of a left translation can be applied to the special embedded matrix Lie groups $SO(n)$, $SE(n)$ and their corresponding lie algebras $\mathfrak{so}(n)$, $\mathfrak{se}(n)$.

Differential of left translation $T_R L_H$ on $SO(n)$ The left translation map in $SO(n)$ is defined as $L_H(R) = HR$ with $H, R \in G := SO(n)$. Applied to eq. (2.1) with $v \in T_R G$, it follows that

$$T_R L_H(v) = Hv.$$

The tangent vector $v \in T_R G$ can be represented by $v = T_e L_R(\omega) = R\omega$ using the left translation map of $\omega \in \mathfrak{g} := \mathfrak{so}(n)$ resulting in

$$T_R L_H(R\omega) = H(R\omega). \tag{2.2}$$

Differential of left translation $T_g L_h$ on $SE(n)$ The left translation map in $SE(n)$ is defined as $L_h(g) = hg$ with $h, g \in G := SE(n)$. Applied to eq. (2.1) with $v \in T_g G$, it follows that

$$T_g L_h(v) = hv.$$

The tangent vector $v \in T_g G$ can be represented by $v = T_e L_g(\xi) = g\xi$ using the left translation map of $\xi \in \mathfrak{g} := \mathfrak{se}(n)$ resulting in

$$T_g L_h(g\xi) = h(g\xi). \tag{2.3}$$

Chapter 3

Continuous Mechanics of Rigid Bodies on Lie Groups

3.1 Lagrangian Mechanics

3.1.1 Euler-Lagrange Equations

$$Euler \tag{3.1}$$

Proposition 2 (Directional derivative representation of $D_\xi L(g,\xi)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L : G \times \mathfrak{g} \to \mathbb{R}$ be smooth. For all $\delta\xi \in \mathfrak{g}$ we regard the partial derivative $D_\xi L(g,\xi) \in \mathfrak{g}^*$ as the covector characterized by*

$$\langle D_\xi L(g,\xi), \delta\xi \rangle = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g, \xi + \varepsilon\,\delta\xi). \tag{3.2}$$

Proof. Since $\mathfrak{g}$ is a vector space, the differential of the map $L(g,\xi)$ at ξ is a linear map $\mathfrak{g} \to \mathbb{R}$, as g is fixed. Therefore the partial derivative $D_\xi L(g,\xi)$ is an element of the dual space $\mathfrak{g}^*$. Moreover, for any $\delta\xi \in \mathfrak{g}$, the curve $\gamma(\varepsilon) = \xi + \varepsilon\,\delta\xi$ satisfies $\gamma(0) = \xi$ and $\dot\gamma(0) = \delta\xi$, which realizes $\delta\xi$ as a tangent direction at ξ for any $\varepsilon \in (-c, c)$ [Lee08]. The corresponding directional derivative is

$$\langle D_\xi L(g,\xi), \delta\xi \rangle = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g, \gamma(\varepsilon)) = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g, \xi + \varepsilon\,\delta\xi),$$

using $\langle D_\xi L(g,\xi), \delta\xi \rangle = D_\xi L(g,\xi)(\delta\xi)$. □

Proposition 3 (Directional derivative representation of $D_g L(g,\xi)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L : G \times \mathfrak{g} \to \mathbb{R}$ be smooth and η an arbitrary curve in $\mathfrak{g}$. For all $\delta g \in T_g G$ we regard the partial derivative $D_g L(g,\xi) \in T_g^* G$ as the covector characterized by*

$$\langle D_g L(g,\xi), \delta g \rangle = \langle T_e^* L_g D_g L(g,\xi), \eta \rangle = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} L(g \exp(\varepsilon\eta), \xi). \tag{3.3}$$

Proof. Since G is a smooth manifold, the differential of the map $L(g,\xi)$ is a linear map $T_gG \to \mathbb{R}$, as ξ is fixed. Therefore the partial derivative $D_gL(g,\xi)$ is an element of the dual space T_g^*G. Moreover, for any $\eta \in \mathfrak{g}$, let $\gamma(\varepsilon) = g\exp(\varepsilon\eta)$ being the smooth curve on the manifold. Then $\gamma(0) = g$ and

$$\begin{aligned}\delta g \;=\; \dot\gamma(0) &= \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0}\gamma(\varepsilon) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} g\exp(\varepsilon\eta)\\ &= g\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0}\exp(\varepsilon\eta) = g\,\eta\\ &= T_eL_g(\eta),\end{aligned} \tag{3.4}$$

with δg being the infinitesimal variation of g induced by η for any $\varepsilon \in (-c,c)$ [Lee08]. The corresponding directional derivative is,

$$\langle D_gL(g,\xi),\delta g\rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(\gamma(\varepsilon),\xi) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L(g\exp(\varepsilon\eta),\xi),$$

using $\langle D_gL(g,\xi),\delta g\rangle = D_gL(g,\xi)(\delta g)$ [Lee08]. Additionally, the directional derivative can be expressed by the pulled-back covector $\mu_g = T_e^*L_gD_gL(g,\xi)$ to

$$\langle D_gL(g,\xi),\delta g\rangle = \langle D_gL(g,\xi), g\eta\rangle = \langle D_gL(g,\xi), T_eL_g(\eta)\rangle = \langle T_e^*L_gD_gL(g,\xi),\eta\rangle,$$

using eq. (3.4). □

3.1.2 Hamiltonian Mechanics

3.1.3 Legendre Transformation

3.1.4 Properties of the Lagrangian Flow

3.2 Unforced 3D-Pendulum

In the following Section, the unforced mechanics of a three-degree-of-freedom pendulum given by [Lee08] are presented, using Lagrangian mechanics of section 3.1. Following, the momentum preservation properties of the Lagrangian flow of the model will be verified.

3.2.1 Derivation of Lie Group Mechanics

Configuration Manifold The 3D-Pendulum model, illustrated in **??**, consist of a rigid body rotating under the influence of gravity g attached to a fixed frictionless pivot [SSC+04]. Moreover, the system has no external control moment, since it is unforced. A space frame and a body-fixed frame attached to the pendulum, with origin at the pivot are defined. The gravity direction in the space frame is represented by the unit vector $\boldsymbol{e}_3$, defining the vector from the pivot to the center of mass to $\boldsymbol{\rho_c}$ in the body frame. Following the fixed pivot, there is no translational degree of freedom. Therefore, the configuration manifold of the model is given by the special orthogonal group

$$SO(3) = \left\{ \boldsymbol{R} \in \mathbb{R}^{3\times 3} \;\middle|\; \boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}_3,\ \det(\boldsymbol{R}) = 1 \right\}.$$

The configuration is completely determined by the rotation matrix $\boldsymbol{R} \in SO(3)$, which maps the body-fixed frame to the space frame. Additionally, in $SO(3)$ the attitude kinematics equation is defined by

$$\dot{\boldsymbol{R}} = \boldsymbol{R}\hat{\boldsymbol{\omega}}, \tag{3.5}$$

with the angular velocity $\boldsymbol{\omega} \in \mathbb{R}^3$, mapped to the Lie algebra $\mathfrak{so}(3)$ by the hat map $\hat{\cdot} : \mathbb{R}^3 \to \mathfrak{so}(3)$ [Lee08]. Consequently, $\hat{\boldsymbol{\omega}}$ is a 3×3 skew symmetric matrix.

Lagrangian To derive the Lagrangian $L : SO(3) \times \mathfrak{so}(3) \to \mathbb{R}$, the kinetic $T : SO(3) \times \mathfrak{so}(3) \to \mathbb{R}$ and potential $U : SO(3) \to \mathbb{R}$ energy of the rigid body are introduced. The rotational inertia of the mass is expressed in the body-frame by the inertia matrix $\boldsymbol{J} \in \mathbb{R}^{3\times 3}$. [Lee08] defines

$$T(\boldsymbol{\omega}) = \int_B \|\boldsymbol{\omega}\hat{\boldsymbol{\rho}}_c\|^2 \, dm(\rho_c) = \frac{1}{2}\operatorname{tr}[\hat{\boldsymbol{\omega}}\boldsymbol{J}_d\hat{\boldsymbol{\omega}}^\top] = \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{J}\boldsymbol{\omega}, \tag{3.6}$$

as a general expression for the kinetic energy for the single body B with the velocity $\boldsymbol{\omega} \times \boldsymbol{\rho}_c = \hat{\boldsymbol{\omega}}\boldsymbol{\rho}_c$, where

$$\boldsymbol{J_d} = \int_B \hat{\boldsymbol{\rho}}_c \hat{\boldsymbol{\rho}}_c^\top \, dm = \frac{1}{2}\operatorname{tr}[\boldsymbol{J}]\,\boldsymbol{I} - \boldsymbol{J}, \tag{3.7}$$

is a nonstandard inertia matrix, with the relation

$$(\boldsymbol{J}\boldsymbol{\omega})^\wedge = \hat{\boldsymbol{\omega}}\boldsymbol{J}_d + \boldsymbol{J}_d\hat{\boldsymbol{\omega}}, \tag{3.8}$$

for any $\boldsymbol{\omega} \in \mathbb{R}^3$. The potential energy is given by

$$U(\boldsymbol{R}) = -mg\, \boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c, \tag{3.9}$$

where $\boldsymbol{R}\boldsymbol{\rho}_c$ denotes the vector from the pivot to the center of mass expressed in the space-fixed frame, and $\boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c$ is its projection onto the gravity direction $\boldsymbol{e}_3$ [Lee08]. Using (3.6) and (3.9), the Lagrangian of the unforced 3D pendulum is given by

$$L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = T(\hat{\boldsymbol{\omega}}) - U(\boldsymbol{R}) = \frac{1}{2}\operatorname{tr}[\hat{\boldsymbol{\omega}}\boldsymbol{J}_d\hat{\boldsymbol{\omega}}^\top] + mg\, \boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c. \tag{3.10}$$

Euler-Lagrange Equations In order to define the Euler-Lagrange Equation (3.1), the derivatives of (3.10) have to be calculated. Firstly, the directional derivative $D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}})$ is given by applying (3.2), which yields

$$\begin{aligned}
\langle D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \delta\hat{\boldsymbol{\omega}} \rangle &= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}} + \epsilon\,\delta\hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \frac{1}{2}\operatorname{tr}\big[(\hat{\boldsymbol{\omega}} + \epsilon\,\delta\hat{\boldsymbol{\omega}})\,\boldsymbol{J}_d\,(\hat{\boldsymbol{\omega}} + \epsilon\,\delta\hat{\boldsymbol{\omega}})^T\big] \\
&= \frac{1}{2}\operatorname{tr}\big[\delta\hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,\hat{\boldsymbol{\omega}}^T + \hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,(\delta\hat{\boldsymbol{\omega}})^T\big]\,.
\end{aligned}$$

By using the antisymmetry identity of a cross product $\hat{\boldsymbol{x}} = -\hat{\boldsymbol{x}}^\top$ for any $\boldsymbol{x} \in \mathbb{R}^3$ and $\operatorname{tr}[\boldsymbol{A}\boldsymbol{B}] = \operatorname{tr}[\boldsymbol{B}\boldsymbol{A}]$ for any $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{3\times3}$, the directional derivative is rewritten as

$$\begin{aligned}
\langle D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \delta\hat{\boldsymbol{\omega}} \rangle &= \frac{1}{2}\operatorname{tr}[-\delta\hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,\hat{\boldsymbol{\omega}} - \hat{\boldsymbol{\omega}}\,\boldsymbol{J}_d\,\delta\hat{\boldsymbol{\omega}}] \\
&= \frac{1}{2}\operatorname{tr}\big[-\big(\boldsymbol{J}_d\hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}}\boldsymbol{J}_d\big)\,\delta\hat{\boldsymbol{\omega}}\big] \\
&= \langle \boldsymbol{J}_d\hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}}\boldsymbol{J}_d,\, \delta\hat{\boldsymbol{\omega}} \rangle \\
&= \langle (\boldsymbol{J}\boldsymbol{\omega})^\wedge,\, \delta\hat{\boldsymbol{\omega}} \rangle\,,
\end{aligned} \tag{3.11}$$

with the definition of the inner product **??** and the skew-symmetric property of (3.8).

Moreover, the derivative $D_{\boldsymbol{R}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}})$, which is in this model only dependent of the potential energy $U(\boldsymbol{R})$, is calculated by (3.3) to

$$\begin{aligned}
\langle D_{\boldsymbol{R}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \delta\boldsymbol{R} \rangle &= \langle T_e^* L_{\boldsymbol{R}}\, D_{\boldsymbol{R}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \hat{\boldsymbol{\eta}} \rangle \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} L(\boldsymbol{R}\exp(\epsilon\hat{\boldsymbol{\eta}}), \hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} mg\,\boldsymbol{e}_3^\top \boldsymbol{R}\exp(\epsilon\hat{\boldsymbol{\eta}})\boldsymbol{\rho}_c \\
&= mg\,\boldsymbol{e}_3^\top \boldsymbol{R}\hat{\boldsymbol{\eta}}\,\boldsymbol{\rho}_c \\
&= mg\,(\boldsymbol{R}^\top\boldsymbol{e}_3)^\top\hat{\boldsymbol{\eta}}\,\boldsymbol{\rho}_c \\
&= mg\,(\boldsymbol{\rho}_c \times \boldsymbol{R}^\top\boldsymbol{e}_3)\cdot\boldsymbol{\eta},
\end{aligned}$$

since $\boldsymbol{a}^\top\hat{\boldsymbol{x}}\boldsymbol{b} = \boldsymbol{a}\cdot(\boldsymbol{x}\times\boldsymbol{b}) = (\boldsymbol{b}\times\boldsymbol{a})\cdot\boldsymbol{x}$ for any $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^3$. Applying the definition of inner products (**??**), it follows that

$$\langle T_e^* L_{\boldsymbol{R}}\, D_{\boldsymbol{R}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \hat{\boldsymbol{\eta}} \rangle = \big\langle (mg\,(\boldsymbol{\rho}_c \times \boldsymbol{R}^\top\boldsymbol{e}_3))^\wedge, \hat{\boldsymbol{\eta}} \big\rangle\,. \tag{3.12}$$

Since the directional derivative

$$D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) \mathrel{\widehat{=}} \boldsymbol{J}_d\hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}}\boldsymbol{J}_d,$$

was derived in (3.11), the coadjoint term in the Euler–Lagrange equations can be written in matrix form by using the identity (**??**). Therefore,

$$\begin{aligned} ad^*_{\hat{\boldsymbol{\omega}}} D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) &= [\boldsymbol{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \boldsymbol{J}_d, \hat{\boldsymbol{\omega}}] \\ &= (\boldsymbol{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \boldsymbol{J}_d) \hat{\boldsymbol{\omega}} - \hat{\boldsymbol{\omega}} (\boldsymbol{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \boldsymbol{J}_d) \\ &= (\boldsymbol{J}\boldsymbol{\omega} \times \boldsymbol{\omega})^\wedge \end{aligned}$$

with the relation $\hat{\boldsymbol{x}}\hat{\boldsymbol{y}} - \hat{\boldsymbol{y}}\hat{\boldsymbol{x}} = (\boldsymbol{x} \times \boldsymbol{y})^\wedge$ for any $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^3$ and (3.8). Furthermore by the antisymmetry of the cross product, it follows that

$$ad^*_{\hat{\boldsymbol{\omega}}} D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = (\boldsymbol{J}\boldsymbol{\omega} \times \boldsymbol{\omega})^\wedge = -(\boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega})^\wedge \tag{3.13}$$

Finally, the Euler-Lagrange equations of the unforced 3D-pendulum are expressed in $\mathfrak{so}(3)$ $SO(3)$ form, by substituting (3.11), (3.12) and (3.13) into (3.1), which result in

$$(\boldsymbol{J}\dot{\boldsymbol{\omega}})^\wedge + (\boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}))^\wedge = (mg\, \boldsymbol{\rho}_c \times \boldsymbol{R}^T \boldsymbol{e}_3)^\wedge, \tag{3.14a}$$

$$\dot{\boldsymbol{R}} = \boldsymbol{R}\hat{\boldsymbol{\omega}}. \tag{3.14b}$$

Legendre Transformation In order to obtain the Hamilton equations for the unforced 3D-pendulum model, the Legendre transformation $\mathbb{F}L : (SO(3) \times \mathfrak{so}(3)) \to (SO(3) \times \mathfrak{so}(3)^*)$ (**??**) has to be applied, which results in the expression for the momentum

$$\hat{\boldsymbol{\Pi}} = D_{\hat{\boldsymbol{\omega}}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = (\boldsymbol{J}\boldsymbol{\omega})^\wedge \tag{3.15}$$

where $\hat{\boldsymbol{\Pi}} \in \mathfrak{so}(3)^*$. By substituting (3.15) into (3.14), the Hamilton equations of an unforced 3D-pendulum are

$$\hat{\dot{\boldsymbol{\Pi}}} + (\boldsymbol{J}^{-1}\boldsymbol{\Pi} \times \boldsymbol{\Pi})^\wedge = (mg\, \boldsymbol{\rho}_c \times \boldsymbol{R}^T \boldsymbol{e}_3)^\wedge, \tag{3.16a}$$

$$\dot{\boldsymbol{R}} = \boldsymbol{R}(\boldsymbol{J}^{-1}\boldsymbol{\Pi})^\wedge. \tag{3.16b}$$

Finally, the derived Euler-Lagrange equations (3.14) and the Hamilton equations (3.16) correspond to the given 3D-pendulum results of [Lee08].

3.2.2 Verification Of Momentum Conservation

The verification of momentum conservation is particularly relevant in the context of this thesis, since the use of Lagrangian mechanics on Lie groups preserves the intrinsic geometric structure of the dynamics, which is essential for accurate simulation and for certifiable trajectory optimization methods [TJVG25]. Therefore, in the following, the momentum conservation property of the continuous 3D pendulum is verified based on its symmetry and Noether's theorem.

[Lee08] identifies a symmetry of the unforced 3D-pendulum's Lagrangian, which is infinitesimally invariant under an action of the group $H = SO(2) \simeq S^1$. Hereby, a rotation about the gravity direction $\boldsymbol{e}_3$ does not change the Lagrangian. Thus, $SO(2)$ and S^1 describe the same symmetry group under the left action of H on G $\Phi : S^1 \times SO(3) \to SO(3)$

$$\Phi(\theta, \boldsymbol{R}) = \exp_{SO(3)}(\theta \hat{\boldsymbol{e}}_3)\boldsymbol{R}, \tag{3.17}$$

where $\exp_{SO(3)}(\theta \hat{\boldsymbol{e}}_3)$ expresses a finite rotation about the gravity direction $\boldsymbol{e}_3$ by the angle θ [Lee08]. By Noether's Theorem **??**, the infinitesimally invariance of the 3D-pendulum's Lagrangian is verified, if

$$dL(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) \cdot \zeta_{SO(3)\times\mathfrak{so}(3)}(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = 0, \tag{3.18}$$

consequently preserving the Lagrangian momentum map along the flow $J_L\big(\mathcal{F}_L^T(\boldsymbol{R}, \hat{\boldsymbol{\omega}})\big) = J_L(\boldsymbol{R}, \hat{\boldsymbol{\omega}})$.

Therefore, the infinitesimal generator of the lifted group action on $SO(3)\times\mathfrak{so}(3)$ is required [Lee08].With the attitude kinematics relation (3.5), the left trivialization $\phi_L(\boldsymbol{R}, \dot{\boldsymbol{R}}) = \phi_L(\boldsymbol{R}, \boldsymbol{R}\hat{\boldsymbol{\omega}}) = (\boldsymbol{R}, T_e L_{\boldsymbol{R}^{-1}} \boldsymbol{R}\hat{\boldsymbol{\omega}}) = (\boldsymbol{R}, \hat{\boldsymbol{\omega}})$, and the left trivialization $\phi_L^{-1}(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = (\boldsymbol{R}, T_e L_{\boldsymbol{R}}\hat{\boldsymbol{\omega}}) = (\boldsymbol{R}, \boldsymbol{R}\hat{\boldsymbol{\omega}})$ are taken from (**??**) and (**??**), with the differential T_R of (2.2). Following [Lee08], the lifted infinitesimal generator $\zeta_{SO(3)\times\mathfrak{so}(3)} : SO(3) \times \mathfrak{so}(3) \to T(SO(3) \times \mathfrak{so}(3))$ is computed to

$$\begin{aligned}
\zeta_{SO(3)\times\mathfrak{so}(3)}(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) &= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \phi_L \circ T_{\boldsymbol{R}}\Phi_{\exp_{SO(2)}(\epsilon\zeta)} \circ \phi_L^{-1}(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \phi_L \circ T_{\boldsymbol{R}}\Phi_{\exp_{SO(2)}(\epsilon\zeta)}(\boldsymbol{R}, \boldsymbol{R}\hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \phi_L(\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\boldsymbol{R}, \exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\boldsymbol{R}\hat{\boldsymbol{\omega}}) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} \big(\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\boldsymbol{R}, \boldsymbol{R}^{-1}\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)^{-1}\big(\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\boldsymbol{R}\hat{\boldsymbol{\omega}}\big)\big) \\
&= \frac{d}{d\epsilon}\bigg|_{\epsilon=0} (\exp(\epsilon\zeta\hat{\boldsymbol{e}}_3)\boldsymbol{R}, \hat{\boldsymbol{\omega}}) \\
&= (\zeta\hat{\boldsymbol{e}}_3\boldsymbol{R}, \boldsymbol{0})\,.
\end{aligned} \tag{3.19}$$

This result can be substituted into (3.18), using (3.12), which gives

$$\begin{aligned}
dL(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) \cdot \zeta_{SO(3)\times\mathfrak{so}(3)}(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) &= dL(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) \cdot (\zeta\hat{\boldsymbol{e}}_3\boldsymbol{R}, \boldsymbol{0}) \\
&= \langle D_{\boldsymbol{R}}L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \zeta\hat{\boldsymbol{e}}_3\boldsymbol{R}\rangle + \langle D_{\hat{\boldsymbol{\omega}}}L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \boldsymbol{0}\rangle \\
&= \langle T_e^* L_{\boldsymbol{R}} D_{\boldsymbol{R}}L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \hat{\boldsymbol{\eta}}\rangle\,.
\end{aligned} \tag{3.20}$$

Since the lifted generator has no component in the $\hat{\boldsymbol{\omega}}$ direction, the component directional derivative $D_{\hat{\boldsymbol{\omega}}}$ equals zero in zero direction. Moreover, by the definition of the lifted infinitesimal generator (**??**), the result of (3.19) redefines the variational relationship being

$$(\delta\boldsymbol{R}, \delta\hat{\boldsymbol{\omega}}) = (\zeta\hat{\boldsymbol{e}}_3\boldsymbol{R}, \boldsymbol{0}). \tag{3.21}$$

In consequence, with the new relation of $\delta\boldsymbol{R} = \zeta\hat{\boldsymbol{e}}_3\boldsymbol{R} = \boldsymbol{R}\hat{\boldsymbol{\eta}}$, the variation $\hat{\boldsymbol{\eta}}$ can be expressed as

$$\hat{\boldsymbol{\eta}} = \zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}. \tag{3.22}$$

This can be substituted into (3.20), following

$$\left\langle T_e^* L_{\boldsymbol{R}} D_{\boldsymbol{R}} L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}),\, \zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}\right\rangle = \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L\big(\boldsymbol{R}\exp\big(\epsilon\zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}\big)\,, \hat{\boldsymbol{\omega}}\big)\,.$$

Since only the potential term of the 3D-pendulum $U(\boldsymbol{R})$ (3.9) depends on $\boldsymbol{R}$ and (3.22) is skew symmetric, it can be shown that

$$\begin{aligned}
\left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L\big(\boldsymbol{R}\exp\big(\epsilon\zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}\big)\,, \hat{\boldsymbol{\omega}}\big) &= mg\, \boldsymbol{e}_3^T\boldsymbol{R}\left(\zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}\right)\boldsymbol{\rho}_c \\
&= mg\left(\boldsymbol{\rho}_c^\top\left(\zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}\right)^\top\boldsymbol{R}^\top\boldsymbol{e}_3\right)^\top \\
&= -\,mg\,\boldsymbol{\rho}_c^\top\left(\zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{R}\right)\boldsymbol{R}^\top\boldsymbol{e}_3 \\
&= -\,mg\,\boldsymbol{\rho}_c^\top\zeta\boldsymbol{R}^{-1}\hat{\boldsymbol{e}}_3\boldsymbol{e}_3 \\
&= -\,mg\,\boldsymbol{\rho}_c^\top\zeta\boldsymbol{R}^{-1}(\boldsymbol{e}_3\times\boldsymbol{e}_3) \\
&= 0,
\end{aligned} \tag{3.23}$$

because $\boldsymbol{e}_3\times\boldsymbol{e}_3 = \mathbf{0}$ and $\boldsymbol{R}^{-1} = \boldsymbol{R}^\top$ for any $\boldsymbol{R}\in SO(3)$. Therefore, the condition of (3.18) is verified and the Lagrangian is infinitesimally invariant under the symmetry action. Consequently, by Noether's Theorem **??**, the corresponding momentum map $J_L(\boldsymbol{R},\boldsymbol{\omega}) = \boldsymbol{e}_3^T\boldsymbol{R}\boldsymbol{J}\boldsymbol{\omega}$ of the continuous 3D-pendulum, which is given in [Lee08], is preserved along the Lagrangian flow

$$J_L\big(\mathcal{F}_L^T(\boldsymbol{R},\hat{\boldsymbol{\omega}})\big) = J_L(\boldsymbol{R},\hat{\boldsymbol{\omega}}).$$

Chapter 4

Discrete Mechanics For Rigid Bodies On Lie Groups

4.1 Discrete Lagrangian Mechanics

4.1.1 Discrete Euler-Lagrange Equations

Proposition 4 (Directional derivative representation of $D_{g_k}L_d(g_k, f_k)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L_d : G \times G \to \mathbb{R}$ be smooth and η_k an arbitrary element of $\mathfrak{g}$. For all $\delta g_k \in T_{g_k}G$ with $\delta g_k = T_e L_{g_k}(\eta_k)$ (**??**), we regard the partial derivative $D_{g_k}L_d(g_k, f_k) \in T^*_{g_k}G$ as the covector characterized by*

$$\langle D_{g_k}L_d(g_k, f_k), \delta g_k\rangle = \langle T^*_e L_{g_k} D_{g_k}L_d(g_k, f_k), \eta_k\rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(g_k \exp(\varepsilon\eta_k), f_k). \tag{4.1}$$

Proof. Since G is a smooth manifold, the differential of the map $L_d(g_k, f_k)$ is a linear map $T_{g_k}G \to \mathbb{R}$, as f_k is fixed. Therefore, the partial derivative $D_{g_k}L_d(g_k, f_k)$ is an element of the dual space $T^*_{g_k}G$. Moreover, for any $\eta_k \in \mathfrak{g}$, let

$$\gamma(\varepsilon) = g_k \exp(\varepsilon\eta_k)$$

be a smooth curve on the manifold. Then $\gamma(0) = g_k$ and

$$\begin{aligned}\delta g_k \;&=\; \dot\gamma(0) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0}\gamma(\varepsilon) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} g_k \exp(\varepsilon\eta_k)\\ &= g_k \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0}\exp(\varepsilon\eta_k) = g_k\,\eta_k\\ &= T_e L_{g_k}(\eta_k),\end{aligned} \tag{4.2}$$

with δg_k being the infinitesimal variation of g_k induced by η_k for any $\varepsilon \in (-c, c)$ [Lee08]. The corresponding directional derivative is

$$\langle D_{g_k}L_d(g_k, f_k), \delta g_k\rangle = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(\gamma(\varepsilon), f_k) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} L_d(g_k \exp(\varepsilon\eta_k), f_k),$$

using $\langle D_{g_k} L_d(g_k, f_k), \delta g_k \rangle = D_{g_k} L_d(g_k, f_k)(\delta g_k)$ [Lee08]. Additionally, the directional derivative can be expressed by the pulled-back covector

$$\mu_k = T_e^* L_{g_k} D_{g_k} L_d(g_k, f_k)$$

to obtain

$$\begin{aligned}
\langle D_{g_k} L_d(g_k, f_k), \delta g_k \rangle &= \langle D_{g_k} L_d(g_k, f_k), g_k \eta_k \rangle \\
&= \langle D_{g_k} L_d(g_k, f_k), T_e L_{g_k}(\eta_k) \rangle \\
&= \langle T_e^* L_{g_k} D_{g_k} L_d(g_k, f_k), \eta_k \rangle,
\end{aligned}$$

using eq. (4.2). □

Proposition 5 (Directional derivative representation of $D_{f_k} L_d(g_k, f_k)$)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$ [Hal00]. Let $L_d : G \times G \to \mathbb{R}$ be smooth and X_k an arbitrary element of $\mathfrak{g}$. For all $\delta f_k \in T_{f_k} G$ with $\delta f_k = T_e L_{f_k}(X_k)$ (**??**), we regard the partial derivative $D_{f_k} L_d(g_k, f_k) \in T^*_{f_k} G$ as the covector characterized by*

$$\langle D_{f_k} L_d(g_k, f_k), \delta f_k \rangle = \langle T_e^* L_{f_k} D_{f_k} L_d(g_k, f_k), X_k \rangle = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L_d(g_k, f_k \exp(\varepsilon X_k)). \tag{4.3}$$

Proof. Since G is a smooth manifold, the differential of the map $L_d(g_k, f_k)$ is a linear map $T_{f_k} G \to \mathbb{R}$, as g_k is fixed. Therefore, the partial derivative $D_{f_k} L_d(g_k, f_k)$ is an element of the dual space $T^*_{f_k} G$. Moreover, for any $X_k \in \mathfrak{g}$, let

$$\gamma(\varepsilon) = f_k \exp(\varepsilon X_k)$$

be a smooth curve on the manifold. Then $\gamma(0) = f_k$ and

$$\begin{aligned}
\delta f_k \;=\; \dot\gamma(0) &= \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \gamma(\varepsilon) = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} f_k \exp(\varepsilon X_k) \\
&= f_k \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \exp(\varepsilon X_k) = f_k X_k \\
&= T_e L_{f_k}(X_k),
\end{aligned} \tag{4.4}$$

with δf_k being the infinitesimal variation of f_k induced by X_k for any $\varepsilon \in (-c, c)$ [Lee08]. The corresponding directional derivative is,

$$\langle D_{f_k} L_d(g_k, f_k), \delta f_k \rangle = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L_d(g_k, \gamma(\varepsilon)) = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L_d(g_k, f_k \exp(\varepsilon X_k)),$$

using $\langle D_{f_k} L_d(g_k, f_k), \delta f_k \rangle = D_{f_k} L_d(g_k, f_k)(\delta f_k)$ [Lee08]. Additionally, the directional derivative can be expressed by the pulled-back covector

$$\mu_k = T_e^* L_{f_k} D_{f_k} L_d(g_k, f_k)$$

to obtain

$$\begin{aligned}\langle D_{f_k} L_d(g_k, f_k), \delta f_k\rangle &= \langle D_{f_k} L_d(g_k, f_k), f_k X_k\rangle \\ &= \langle D_{f_k} L_d(g_k, f_k), T_e L_{f_k}(X_k)\rangle \\ &= \langle T_e^* L_{f_k} D_{f_k} L_d(g_k, f_k), X_k\rangle,\end{aligned}$$

using eq. (4.4). □

4.1.2 Discrete Hamiltonian Mechanics

4.1.3 Discrete Legendre Transformation

4.1.4 Properties of the Discrete Lagrangian Flow

4.1.5 Forced Discrete Euler-Lagrange Equations

Currently, the discrete Euler-Lagrange equations (**??**) describe unforced mechanical systems. However, in most cases the system is subjected to external forces or control inputs. The Lagrange-d'Alembert principle can be applied to derive the forced Euler-Lagrange equations for continuous mechanics, given by

$$\delta \int_{t_0}^{t_f} L(\boldsymbol{g}, \boldsymbol{\xi})\, dt + \int_{t_0}^{t_f} \boldsymbol{u}(t) \cdot \boldsymbol{\eta}\, dt = 0, \tag{4.5}$$

with generalized forces $\boldsymbol{u}(t) : [t_0, t_f] \to \mathfrak{g}$ acting on the system for any $\boldsymbol{\eta} = \boldsymbol{g^{-1}\delta g} \in \mathfrak{g}$ in continuous time [Lee08].

In discrete time the action integral of the Lagrangian is approximated by

$$\int_{t_0}^{t_f} \boldsymbol{u}(t) \cdot \boldsymbol{\eta}\, dt \approx \boldsymbol{u}_{d_k}^- \cdot \boldsymbol{\eta}_k + \boldsymbol{u}_{d_k}^+ \cdot \boldsymbol{\eta}_{k+1}, \tag{4.6}$$

using discrete generalized forces $\boldsymbol{u}_{d_k}^+, \boldsymbol{u}_{d_k}^- \in \mathfrak{g}$ acting on the system at the discrete time step k [MW01]. Therefore, corresponding to the action sum of the discrete Lagrangian (**??**), the discrete Lagrange-d'Alembert principle is given by

$$\delta \sum_{k=0}^{N-1} L_d(\boldsymbol{g}_k, \boldsymbol{f}_k) + \sum_{k=0}^{N-1} \left(\boldsymbol{u}_{d_k}^- \cdot \boldsymbol{\eta}_k + \boldsymbol{u}_{d_k}^+ \cdot \boldsymbol{\eta}_{k+1}\right) = 0. \tag{4.7}$$

[Lee08] states that (4.7) is equivalent to the forced discrete Euler-Lagrange equations (**??**) with additional forcing term included. In consequence, the forced discrete Euler-Lagrange equations are given by

$$\begin{aligned}&\boldsymbol{T_e^* L_{f_{k-1}}} \cdot \boldsymbol{D_{f_{k-1}} L}_d(\boldsymbol{g}_{k-1}, \boldsymbol{f}_{k-1}) - \boldsymbol{Ad^*_{f_{k-1}}} \cdot (\boldsymbol{T_e^* L_{f_k}} \cdot \boldsymbol{D_{f_k} L}_d(\boldsymbol{g}_k, \boldsymbol{f}_k)) \\ &\quad + \boldsymbol{T_e^* L_{g_k}} \cdot \boldsymbol{D_{g_k} L}_d(\boldsymbol{g}_k, \boldsymbol{f}_k) + \boldsymbol{u_{d_k}^-} + \boldsymbol{u_{d_{k-1}}^+} = \boldsymbol{0},\end{aligned} \tag{4.8a}$$

$$\boldsymbol{g}_{k+1} = \boldsymbol{g}_k \boldsymbol{f}_k. \tag{4.8b}$$

For discrete Lagrangian of the form

$$L_d(\boldsymbol{g}_k, \boldsymbol{f}_k) = T_d(\boldsymbol{f}_k) - (1-c)hU(\boldsymbol{g}_k) - chU(\boldsymbol{g}_k \boldsymbol{f}_k), \tag{4.9}$$

with $T_d(\boldsymbol{f}_k)$ being the discrete kinetic energy, $U(\boldsymbol{g}_k)$ the potential energy and $c \in [0,1]$ a constant parameter, the forced discrete Euler-Lagrange equations can be written in a more compact form [Lee08]. For this formulation, the continuous control inputs are parametrized by their values at the discrete time nodes, and the corresponding discrete generalized forces are chosen as

$$\boldsymbol{u}^-_{d_k} = (1-c)h\boldsymbol{u}_k, \qquad \boldsymbol{u}^+_{d_{k-1}} = ch\boldsymbol{u}_k. \tag{4.10}$$

For the mechanical discrete Lagrangian in (4.9), the contribution of the potential energy can be written in terms of the configuration-dependent moment

$$\boldsymbol{M}(\boldsymbol{g}_k) = -\boldsymbol{T}^*_{\boldsymbol{e}}\boldsymbol{L}_{\boldsymbol{g}_k} \cdot \boldsymbol{D}U(\boldsymbol{g}_k). \tag{4.11}$$

Substituting the special discrete moment (4.11) and the discrete generalized forces from (4.10) into the forced discrete Euler-Lagrange (4.8) equations yields

$$\begin{aligned} \boldsymbol{T}^*_{\boldsymbol{e}}\boldsymbol{L}_{\boldsymbol{f}_{k-1}} \cdot \boldsymbol{D}_{\boldsymbol{f}_{k-1}} T_d(\boldsymbol{f}_{k-1}) - \boldsymbol{Ad}^*_{\boldsymbol{f}_k^{-1}} \cdot (\boldsymbol{T}^*_{\boldsymbol{e}}\boldsymbol{L}_{\boldsymbol{f}_k} \cdot \boldsymbol{D}_{\boldsymbol{f}_k} T_d(\boldsymbol{f}_k)) \\ + h\boldsymbol{M}(\boldsymbol{g}_k) + h\boldsymbol{u}_k = \boldsymbol{0}. \end{aligned} \tag{4.12a}$$

$$\boldsymbol{g}_{k+1} = \boldsymbol{g}_k \boldsymbol{f}_k. \tag{4.12b}$$

Compared to the unforced discrete Euler-Lagrange equations (**??**), the forced equations include the additional control term $h\boldsymbol{u}_k$.

Using the discrete Legendre transformation (**??**), the same dynamics can also be written in an equivalent momentum form. Then the forced discrete-time Hamiltonian equations are given by

$$\boldsymbol{\Pi}_k = \boldsymbol{Ad}^*_{\boldsymbol{f}_k^{-1}} \cdot (\boldsymbol{T}^*_{\boldsymbol{e}}\boldsymbol{L}_{\boldsymbol{f}_k} \cdot \boldsymbol{D}_{\boldsymbol{f}_k} T_d(\boldsymbol{f}_k)) - (1-c)h\boldsymbol{M}(\boldsymbol{g}_k) - (1-c)h\boldsymbol{u}_k, \tag{4.13a}$$

$$\boldsymbol{g}_{k+1} = \boldsymbol{g}_k \boldsymbol{f}_k. \tag{4.13b}$$

$$\boldsymbol{\Pi}_{k+1} = \boldsymbol{Ad}^*_{\boldsymbol{f}_k} \cdot (\boldsymbol{\Pi}_k + (1-c)h\boldsymbol{M}(\boldsymbol{g}_k) + (1-c)h\boldsymbol{u}_k) + ch\boldsymbol{M}(\boldsymbol{g}_{k+1}) + ch\boldsymbol{u}_{k+1}, \tag{4.13c}$$

This form separates the evolution of the discrete momentum from the reconstruction of the group element. It is particularly useful for numerical simulations, where the control input acts directly on the momentum equation [Lee08].

4.1.6 Computational Implementation

In contrast to the continuous mechanics of **??**, numerical simulations can be performed for discrete-time mechanical model representations. Therefore, a simulation framework has to be defined. To determine the relative attitude update $\boldsymbol{F}_k \in SO(3)$, the implicit Euler-Lagrange (**??**) and Hamilton equations (**??**) have to be solved at every time step k. Therefore, an efficient computational algorithm is required to for practical numerical simulations, especially for complex multibody-mechanical models. [Lee08] proposes an approach of expressing the group element $\boldsymbol{F}_k \in G$ on the linear vector space of the Lie algebra, by using the exponential map. Thereby, the implicit equation is solved numerically by a Newton iteration [MW01] in $\mathbb{R}^3$.

Firstly, the implicit discrete-time Euler-Lagrange equations or the discrete-time Hamilton equations have to be converted into an equivalent vector equation in $\mathbb{R}^3$, for example, using the Rodrigures formula

$$\boldsymbol{F} = \exp(\hat{\boldsymbol{f}}) = \boldsymbol{I}_{3\times 3} + \frac{\sin\|\boldsymbol{f}\|}{\|\boldsymbol{f}\|}\,\hat{\boldsymbol{f}} + \frac{1-\cos\|\boldsymbol{f}\|}{\|\boldsymbol{f}\|^2}\,\hat{\boldsymbol{f}}^{\,2}, \tag{4.14}$$

for $f \in \mathbb{R}^3$ [MR99]. However, it requires the evaluation of expensive sin and cos terms, which increase the computational time and costs. In order to improve the computational efficiency, the exponential map (4.14) is replaced by the Cayley transformation [Ise01].

Cayley Transformation The Cayley transformation is a local diffeomorphism between $SO(3) \to \mathbb{R}^3$, which is defined as

$$F = \mathrm{Cay}(f) = (\boldsymbol{I} + \hat{\boldsymbol{f}})(\boldsymbol{I} - \hat{\boldsymbol{f}})^{-1} = \frac{1}{1+\boldsymbol{f}^\top\boldsymbol{f}}((1-\boldsymbol{f}^\top\boldsymbol{f})\boldsymbol{I} + 2\hat{\boldsymbol{f}} + 2\boldsymbol{f}\boldsymbol{f}^\top, \tag{4.15}$$

representing an rotation θ given by $\|f\| = \tan\left(\frac{\theta}{2}\right)$ [BLdD23]. Therefore, the Cayley transformation consists of the constraint $\theta < \pi$, which requires the step size h to be chosen sufficiently small. After substituting the Cayley transformation (4.15) into the implicit equations (**??**) or (**??**), a equivalent vector equation is obtained

$$0 = A(\boldsymbol{f}), \tag{4.16}$$

where $A : \mathbb{R}^3 \to \mathbb{R}^3$ [CJ12].

Newton Iteration For solving the implicit vector equation (4.16) Newton's method is used [MW01]. The iteration rule is given by

$$\boldsymbol{f}^{(i+1)} = \boldsymbol{f}^{(i)} - \nabla A\big(\boldsymbol{f}^{(i)}\big)^{-1} A\big(\boldsymbol{f}^{(i)}\big)\,, \tag{4.17}$$

where convergence to the solution is reached until $\left\|f^{(i+1)} - f^{(i)}\right\| < \epsilon$ for a defined ϵ. A initial guess $\boldsymbol{f}^0$ can be obtained by linearizing the vector equation (4.16) [Lee08]. Finally, the rotation matrix $\boldsymbol{F}_k \in SO(3)$ at k is calculated using the Cayley transformation (4.15).

4.2 Single Rigid Body Systems

4.2.1 3D-Pendulum

In the following Section, the unforced mechanics of a three-degree-of-freedom pendulum given by [Lee08] are presented, using discrete Lagrangian mechanics of section

4.1. This will correspond to the equivalent discrete case of the continuous unforced 3D-pendulum of section 3.2. Following, numerical simulations are performed to verify the momentum conservation and constant energy of the derived mechanical systems equations.

Configuration Manifold The configuration manifold of the unforced 3D-pendulum in a discrete case, consists equal settings of the pendulums space and body-fixed frame. The gravity direction in the space frame is represented by the unit vector $\boldsymbol{e}_3$, defining the vector from the pivot to the center of mass to $\boldsymbol{\rho_c} = \frac{l}{2}\boldsymbol{e}_3^b$ in the body frame. As in the continuous formulation of Section 3.2, the discrete configuration manifold is $SO(3)$. Therefore, the attitude of the 3D-pendulum is described by a rotation matrix $\boldsymbol{R}_k \in SO(3)$, which maps the body-fixed frame to the reference frame at the discrete time step k. [Lee08] defines the discrete-time attitude kinematics equation as

$$\boldsymbol{R}_{k+1} = \boldsymbol{R}_k \boldsymbol{F}_k, \tag{4.18}$$

with the rotation matrix $\boldsymbol{F}_k$, representing the relative attitude update between two integration steps. Additionally, [Lee08] introduces adjoint operator relations for $\boldsymbol{F} \in SO(3)$ and the skew-symmetric matrix $\hat{\boldsymbol{\eta}} \in \mathfrak{so}(3)$, which are given as

$$\mathrm{Ad}_{\boldsymbol{F}}\, \hat{\boldsymbol{\eta}} = \boldsymbol{F}\, \hat{\boldsymbol{\eta}}\, \boldsymbol{F}^\top = (\boldsymbol{F}\boldsymbol{\eta})^\wedge, \tag{4.19}$$

$$\mathrm{Ad}^*_{\boldsymbol{F}}\, \hat{\boldsymbol{\eta}} = \boldsymbol{F}^\top\, \hat{\boldsymbol{\eta}}\, \boldsymbol{F} = (\boldsymbol{F}^\top\boldsymbol{\eta})^\wedge. \tag{4.20}$$

Discrete Lagrangian As introduced in (**??**), the discrete Lagrangian $L_d(\boldsymbol{R}_k, \boldsymbol{F}_k) : SO(3) \times SO(3) \to \mathbb{R}$ is an approximation of the action integral of the continuous Lagrangian over one time step h. In order to discretize the given continuous Lagrangian of (3.10)

$$L(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) = T(\boldsymbol{R}, \hat{\boldsymbol{\omega}}) - U(\boldsymbol{R}) = \frac{1}{2}\,\mathrm{tr}[\hat{\boldsymbol{\omega}} \boldsymbol{J}_d \hat{\boldsymbol{\omega}}^\top] + mg\, \boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c, \tag{4.21}$$

the time dependent angular velocity $\hat{\boldsymbol{\omega}}(t)$ and rotation matrix $\boldsymbol{R}(t)$ have to be represented on the discrete time interval $[t_k, t_{k+1}]$ [MW01]. Given the definition of the exponential for an $n \times n$ matrix $\boldsymbol{X}$

$$\exp(\boldsymbol{X}) = \sum_{m=0}^{\infty} \frac{\boldsymbol{X}^m}{m},$$

[Hal00], its first-order expansion can be written as

$$\exp(\boldsymbol{X}) = \boldsymbol{I} + \boldsymbol{X} + O(\boldsymbol{X}^2). \tag{4.22}$$

Therefore, for small h, the exponential map $\boldsymbol{F}_k = \exp(h\hat{\boldsymbol{\omega}}_k)$ can be approximated using (4.22) to

$$\boldsymbol{F}_k = \exp(h\hat{\boldsymbol{\omega}}_k) \approx \boldsymbol{I} + h\hat{\boldsymbol{\omega}}_k,$$

following the general discrete approximation of the angular velocity

$$\hat{\boldsymbol{\omega}}_k \approx \frac{1}{h}(\boldsymbol{F}_k - \boldsymbol{I}), \tag{4.23}$$

[Lee08]. Secondly, quadrature rules, for example, the midpoint rule and the trapezoidal rule can be applied to create an integrator $F^{L_d} : (\boldsymbol{R}_0, \boldsymbol{F}_0) \rightarrow (\boldsymbol{R}_1, \boldsymbol{F}_1)$, or the symplectic integration method of Strömer-Verlet [MW01]. In order to align with [Lee08], the trapezodial rule is applied for the approximation for an integral, given by

$$\int_{t_k}^{t_{k+1}} f(t)\, dt \approx \frac{h}{2}\left(f(t_k) + f(t_{k+1})\right), \tag{4.24}$$

for $h = t_{k+1} - t_k$ [But16]. Applying (4.24) to the Lagrangian (4.21), it follows

$$\begin{aligned} \int_{t_k}^{t_{k+1}} U(\boldsymbol{R})\, dt = \int_{t_k}^{t_{k+1}} mg\, \boldsymbol{e}_3^\top \boldsymbol{R}\boldsymbol{\rho}_c\, dt &\approx \frac{h}{2}\, mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_k\boldsymbol{\rho}_c + \frac{h}{2}\, mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_{k+1}\boldsymbol{\rho}_c \\ &= \frac{h}{2}\, mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_k\boldsymbol{\rho}_c + \frac{h}{2}\, mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_k\boldsymbol{F}_k\boldsymbol{\rho}_c, \\ &= U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k), \end{aligned} \tag{4.25}$$

using (4.18), as only the potential energy $U(\boldsymbol{R})$ of the continuous Lagrangian depends on $\boldsymbol{R}$. Substituting (4.23) and (4.25) into (4.21), the discrete Lagrangian is calculated to

$$\begin{aligned} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k) &= \frac{1}{2h}\operatorname{tr}\left[(\boldsymbol{F}_k - \boldsymbol{I})\boldsymbol{J}_d(\boldsymbol{F}_k - \boldsymbol{I})^\top\right] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right) \\ &= \frac{1}{2h}\operatorname{tr}\left[\boldsymbol{F}_k\boldsymbol{J}_d\boldsymbol{F}_k^\top - \boldsymbol{J}_d\boldsymbol{F}_k^\top - \boldsymbol{F}_k\boldsymbol{J}_d + \boldsymbol{J}_d\right] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right) \\ &= \frac{1}{2h}\operatorname{tr}[2\boldsymbol{J}_d - 2\boldsymbol{F}_k\boldsymbol{J}_d] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right) \\ &= \frac{1}{h}\operatorname{tr}[(\boldsymbol{I} - \boldsymbol{F}_k)\boldsymbol{J}_d] - \frac{h}{2}\left(U(\boldsymbol{R}_k) + U(\boldsymbol{R}_k\boldsymbol{F}_k)\right), \end{aligned} \tag{4.26}$$

using $\operatorname{tr}[\boldsymbol{F}_k\boldsymbol{J}_d\boldsymbol{F}_k^\top] = \operatorname{tr}[\boldsymbol{J}_d\boldsymbol{F}_k^\top\boldsymbol{F}_k] = \operatorname{tr}[\boldsymbol{J}_d]$ and $\operatorname{tr}[\boldsymbol{J}_d\boldsymbol{F}_k^\top] = \operatorname{tr}[\boldsymbol{F}_k\boldsymbol{J}_d]$, since $\boldsymbol{J}_d$ is defined as an diagonal matrix $\boldsymbol{J}_d = \boldsymbol{J}_d^\top$ [Lee08].

Discrete Euler-Lagrange Equations In order to define the Euler-Lagrange Equation (**??**), the derivatives of (4.21) have to be calculated. Firstly, the directional derivative $\langle D_{\boldsymbol{F_k}} L(\boldsymbol{R}_k, \boldsymbol{F}_k), \delta\boldsymbol{f}_k\rangle = \left\langle T_e^* L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k), \hat{\boldsymbol{X}}_k\right\rangle$ is given by applying (4.3), which yields

$$\begin{aligned} \left\langle T_e^* L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k), \hat{\boldsymbol{X}}_k\right\rangle &= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L_d\Big(\boldsymbol{R}_k, \boldsymbol{F}_k\exp(\epsilon\hat{\boldsymbol{X}}_k)\Big) \\ &= -\frac{1}{h}\operatorname{tr}\Big[\boldsymbol{F}_k\hat{\boldsymbol{X}}_k\boldsymbol{J}_d\Big] + \frac{h}{2} mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_k\boldsymbol{F}_k\hat{\boldsymbol{X}}_k\boldsymbol{\rho}_c \\ &= -\frac{1}{h}\operatorname{tr}\Big[\hat{\boldsymbol{X}}_k\boldsymbol{J}_d\boldsymbol{F}_k\Big] + \frac{h}{2} mg\, \Big((\boldsymbol{R}_k\boldsymbol{F}_k)^\top\boldsymbol{e}_3\Big)^\top \hat{\boldsymbol{X}}_k\boldsymbol{\rho}_c, \end{aligned}$$

using the trace identity $\mathrm{tr}[\boldsymbol{ABC}] = \mathrm{tr}[\boldsymbol{BCA}]$ for any $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{R}^{n\times n}$. Additionally, [Lee08] introduces the relation $\mathrm{tr}[\hat{\boldsymbol{s}}\boldsymbol{B}] = -\left\langle \boldsymbol{B} - \boldsymbol{B}^\top, \hat{\boldsymbol{s}} \right\rangle$ for any $\boldsymbol{s} \in \mathbb{R}^3, \boldsymbol{B} \in \mathbb{R}^{3\times 3}$. Applying it together with $\boldsymbol{a}^\top \hat{\boldsymbol{x}} \boldsymbol{b} = (\boldsymbol{b} \times \boldsymbol{a}) \cdot \boldsymbol{x}$ for any $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{x} \in \mathbb{R}^3$, the inner product defeinition (**??**), and the linearity of the pairing $\langle \boldsymbol{A}, \boldsymbol{B} \rangle + \langle \boldsymbol{C}, \boldsymbol{B} \rangle = \langle \boldsymbol{A} + \boldsymbol{C}, \boldsymbol{B} \rangle$ for any $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{R}^{n\times n}$, the final directional derivative of the discrete Lagrangian respect to $\boldsymbol{F}_k$ is calculated to

$$\begin{aligned}\left\langle T_e^* L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k), \hat{\boldsymbol{X}}_k \right\rangle &= \left\langle \frac{1}{h}\left(\boldsymbol{J}_d \boldsymbol{F}_k - \boldsymbol{F}_k^\top \boldsymbol{J}_d\right), \hat{\boldsymbol{X}}_k \right\rangle + \left\langle \frac{h}{2} mg \left(\boldsymbol{\rho}_c \times (\boldsymbol{R}_k \boldsymbol{F}_k)^\top \boldsymbol{e}_3\right)^\wedge, \hat{\boldsymbol{X}}_k \right\rangle \\ &= \left\langle \frac{1}{h}\left(\boldsymbol{J}_d \boldsymbol{F}_k - \boldsymbol{F}_k^\top \boldsymbol{J}_d\right) + \left(\frac{h}{2} mg\, \boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3\right)^\wedge, \hat{\boldsymbol{X}}_k \right\rangle.\end{aligned}$$

Therefore,

$$T_e^* L_{\boldsymbol{F}_k}\, D_{\boldsymbol{F}_k} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k) \mathrel{\widehat{=}} \frac{1}{h}\left(\boldsymbol{J}_d \boldsymbol{F}_k - \boldsymbol{F}_k^\top \boldsymbol{J}_d\right) + \frac{h}{2} \hat{\boldsymbol{M}}_{k+1}, \tag{4.27}$$

with the definition of

$$\boldsymbol{M}_k = mg\, \boldsymbol{\rho}_c \times \boldsymbol{R}_k^\top \boldsymbol{e}_3, \tag{4.28}$$

being the moment due to the potential energy in body frame [Lee08].

Moreover, the directional derivative

$$\left\langle D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \delta \boldsymbol{R}_{k+1} \right\rangle = \left\langle T_e^* L_{\boldsymbol{R}_{k+1}}\, D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle,$$

which is in this model only dependent of the potential energy $U(\boldsymbol{R}_{k+1}) + U(\boldsymbol{R}_{k+1}\boldsymbol{F}_{k+1})$, is calculated by (4.1) to

$$\begin{aligned}\left\langle T_e^* L_{\boldsymbol{R}_{k+1}}\, D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle &= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} L_d(\boldsymbol{R}_{k+1} \exp(\epsilon \hat{\boldsymbol{\eta}}_{k+1}), \boldsymbol{F}_{k+1}) \\ &= \frac{h}{2} mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_{k+1} \hat{\boldsymbol{\eta}}_{k+1} \boldsymbol{\rho}_c + \frac{h}{2} mg\, \boldsymbol{e}_3^\top \boldsymbol{R}_{k+1} \hat{\boldsymbol{\eta}}_{k+1} \boldsymbol{F}_{k+1} \boldsymbol{\rho}_c \\ &= \frac{h}{2} mg \left[\left(\boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3\right) + \left(\boldsymbol{F}_{k+1} \boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3\right)\right] \cdot \boldsymbol{\eta}_{k+1}\end{aligned}$$

Applying the definition of inner products (**??**), it follows

$$\begin{aligned}\left\langle T_e^* L_{\boldsymbol{R}_{k+1}}\, D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle &= \left\langle \frac{h}{2} mg \left[\left(\boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3\right) + \left(\boldsymbol{F}_{k+1} \boldsymbol{\rho}_c \times \boldsymbol{R}_{k+1}^\top \boldsymbol{e}_3\right)\right]^\wedge, \hat{\boldsymbol{\eta}}_{k+1} \right\rangle. \\ &= \left\langle \frac{h}{2}\left(\hat{\boldsymbol{M}}_{k+1} + (\boldsymbol{F}_{k+1} \boldsymbol{M}_{k+2})^\wedge\right), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle,\end{aligned}$$

which implies that the derivative of (4.21) with respect to $\boldsymbol{R}_{k+1}$ is given to

$$D_{\boldsymbol{R}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1}) = \frac{h}{2}\left[\hat{\boldsymbol{M}}_{k+1} + (\boldsymbol{F}_{k+1} \boldsymbol{M}_{k+2})^\wedge\right]. \tag{4.29}$$

Since the directional derivative $T_e^* L_{\boldsymbol{F}_k} D_{\boldsymbol{F}_k} L_d(\boldsymbol{R}_k, \boldsymbol{F}_k)$ was derived in (4.27), the coadjoint term in the discrete Euler-Lagrange equation (**??**) can be written in matrix form to

$$\begin{aligned}\mathrm{Ad}^*_{\boldsymbol{F}_{k+1}^{-1}}\left(T_e^* L_{\boldsymbol{F}_{k+1}} D_{\boldsymbol{F}_{k+1}} L_d(\boldsymbol{R}_{k+1}, \boldsymbol{F}_{k+1})\right) &= \boldsymbol{F}_{k+1}\left[\frac{1}{h}\left(\boldsymbol{J}_d \boldsymbol{F}_{k+1} - \boldsymbol{F}_{k+1}^\top \boldsymbol{J}_d\right) + \frac{h}{2}\hat{\boldsymbol{M}}_{k+2}\right]\boldsymbol{F}_{k+1}^\top \\ &= \frac{1}{h}\left(\boldsymbol{F}_{k+1}\boldsymbol{J}_d - \boldsymbol{J}_d \boldsymbol{F}_{k+1}^\top\right) + \frac{h}{2}\boldsymbol{F}_{k+1}\hat{\boldsymbol{M}}_{k+2}\boldsymbol{F}_{k+1}^\top \\ &= \frac{1}{h}\left(\boldsymbol{F}_{k+1}\boldsymbol{J}_d - \boldsymbol{J}_d \boldsymbol{F}_{k+1}^\top\right) + \frac{h}{2}(\boldsymbol{F}_{k+1}\boldsymbol{M}_{k+2})^\wedge,\end{aligned} \tag{4.30}$$

by using the identity (4.20).

Finally, the discrete Euler-Lagrange equations of the unforced 3D-pendulum are expressed in $\mathfrak{so}(3)$ $SO(3)$ form, substituting (4.27), (4.29) and (4.30) into (**??**), which results in

$$\frac{1}{h}\left(\boldsymbol{F}_{k+1}\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_{k+1}^\top - \boldsymbol{J}_d\boldsymbol{F}_k + \boldsymbol{F}_k^\top\boldsymbol{J}_d\right) = h\,\hat{\boldsymbol{M}}_{k+1}, \tag{4.31a}$$

$$\boldsymbol{R}_{k+1} = \boldsymbol{R}_k\boldsymbol{F}_k. \tag{4.31b}$$

[Lee08] describes the procedure of obtaining the discrete-time flow map $(\boldsymbol{R}_k, \boldsymbol{R}_{k+1}) \to (\boldsymbol{R}_{k+1}, \boldsymbol{R}_{k+2})$, by solving (4.31a) finding $\boldsymbol{F}_{k+1}$ for given $(\boldsymbol{R}_k, \boldsymbol{R}_{k+1})$ and $\boldsymbol{F}_k = \boldsymbol{R}_k^\top\boldsymbol{R}_{k+1}$. Secondly, $\boldsymbol{R}_{k+2}$ is derived from (4.31b).

Discrete Legendre Transformation In order to obtain the discrete Hamilton equations for the unforced 3D-pendulum model, the discrete Legendre transformation $\mathbb{F}^+L_d, \mathbb{F}^-L_d : (SO(3)\times SO(3)) \to (SO(3)\times\mathfrak{so}(3)^*)$ with $\mathbb{F}^+L_d(\boldsymbol{R}_{k-1}, \boldsymbol{F}_{k-1}) = \mathbb{F}^-L_d(\boldsymbol{R}_k, \boldsymbol{F}_k)$ (**??**) have to be applied, which results in the expression for the discrete momentum

$$\begin{aligned}\hat{\boldsymbol{\Pi}}_k &= -\frac{h}{2}\left(\hat{\boldsymbol{M}}_k + (\boldsymbol{F}_k\boldsymbol{M}_{k+1})^\wedge\right) + \frac{1}{h}\left(\boldsymbol{F}_k\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_k^\top\right) + \frac{h}{2}\left(\boldsymbol{F}_k\boldsymbol{M}_{k+1}\right)^\wedge \\ &= \frac{1}{h}\left(\boldsymbol{F}_k\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_k^\top\right) - \frac{h}{2}\hat{\boldsymbol{M}}_k,\end{aligned} \tag{4.32a}$$

$$\hat{\boldsymbol{\Pi}}_{k+1} = \frac{1}{h}\left(\boldsymbol{J}_d\boldsymbol{F}_k - \boldsymbol{F}_k^\top\boldsymbol{J}_d\right) + \frac{h}{2}\hat{\boldsymbol{M}}_{k+1}. \tag{4.32b}$$

where $\hat{\boldsymbol{\Pi}} \in \mathfrak{so}(3)^*$. Either by substituting (4.32) into (4.31), or by applying (**??**), the discrete Hamilton equations of an unforced 3D-pendulum are calculated to

$$\frac{1}{h}\left(\boldsymbol{F}_k\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_k^\top\right) = \hat{\boldsymbol{\Pi}}_k + \frac{h}{2}\hat{\boldsymbol{M}}_k, \tag{4.33a}$$

$$\boldsymbol{R}_{k+1} = \boldsymbol{R}_k\boldsymbol{F}_k, \tag{4.33b}$$

$$\hat{\boldsymbol{\Pi}}_{k+1} = (\boldsymbol{F}_k^\top\boldsymbol{\Pi}_k)^\wedge + \frac{h}{2}(\boldsymbol{F}_k^\top\boldsymbol{M}_k)^\wedge + \frac{h}{2}\hat{\boldsymbol{M}}_{k+1}. \tag{4.33c}$$

Finally, the discrete map $(\boldsymbol{R}_k, \boldsymbol{\Pi}_k) \to (\boldsymbol{R}_{k+1}, \boldsymbol{\Pi}_{k+1})$ is obtained by implicitly solving the equation (4.33a) for $\boldsymbol{F}_k$ with given $(\boldsymbol{R}_k, \boldsymbol{\Pi}_k)$. In consequence, $\boldsymbol{R}_{k+1}$ is derived from (4.33b), and $\boldsymbol{\Pi}_{k+1}$ from (4.33c).

Controlled 3D Pendulum The derived Euler-Lagrange and Hamilton equations are describing the free motion of the 3D pendulum under the influence of gravity. In order to formulate a control problem, an external control moment $\boldsymbol{u}_k$ is applied to the system. [Lee08] identifies

$$\boldsymbol{u}_k = \boldsymbol{R}_k^\top \boldsymbol{e}_3 \times \boldsymbol{u}_{p_k}, \tag{4.34}$$

in the body fixed frame for a control parameter $\boldsymbol{u}_{p_k} \in \mathbb{R}^3$. Thereby, the external control moment has no component along the gravity direction, since the vector $\boldsymbol{R}_k^\top \boldsymbol{e}_3$ is aligned with the gravity vector. Thus, the control input preserves the angular momentum about the gravity direction.

For the discrete Lagrangian in (4.21), the potential energy is evaluated at both endpoints of the time interval. Comparing its coefficients with the generalized mechanical discrete Lagrangian (4.9) gives $c = \frac{1}{2}$, which corresponds to the trapezoidal approximation of the potential contribution. Consequently, the discrete generalized forces are chosen as

$$\boldsymbol{u}_{d_k}^- = \frac{h}{2}\boldsymbol{u}_k, \qquad \boldsymbol{u}_{d_{k-1}}^+ = \frac{h}{2}\boldsymbol{u}_k. \tag{4.35}$$

In consequence, the controlled 3D-pendulum Euler-Lagrange equations are derived to

$$\frac{1}{h}\left(\boldsymbol{F}_{k+1}\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_{k+1}^\top - \boldsymbol{J}_d\boldsymbol{F}_k + \boldsymbol{F}_k^\top\boldsymbol{J}_d\right) = h\left(\boldsymbol{M}_{k+1} + \boldsymbol{u}_{k+1}\right)^\wedge, \tag{4.36a}$$

$$\boldsymbol{R}_{k+1} = \boldsymbol{R}_k\boldsymbol{F}_k. \tag{4.36b}$$

by substituting the generalized forces (4.35) into the previously derived discrete Euler-Lagrange equations of the uncontrolled 3D-pendulum (4.31), taking into account the Lagrange-d' Alembert principle for forced discrete mechanics (4.8). Respectively, substituting (4.35) and (4.33) into (4.13) the controlled 3D-pendulum Hamilton equations are calculated to

$$\hat{\boldsymbol{\Pi}}_k = \frac{1}{h}\left(\boldsymbol{F}_k\boldsymbol{J}_d - \boldsymbol{J}_d\boldsymbol{F}_k^\top\right) - \frac{h}{2}\hat{\boldsymbol{M}}_k - \frac{h}{2}\hat{\boldsymbol{u}}_k, \tag{4.37a}$$

$$\boldsymbol{R}_{k+1} = \boldsymbol{R}_k\boldsymbol{F}_k, \tag{4.37b}$$

$$\hat{\boldsymbol{\Pi}}_{k+1} = (\boldsymbol{F}_k^\top\boldsymbol{\Pi}_k)^\wedge + \frac{h}{2}\left(\boldsymbol{F}_k^\top(\boldsymbol{M}_k + \boldsymbol{u}_k)\right)^\wedge + \frac{h}{2}(\boldsymbol{M}_{k+1} + \boldsymbol{u}_{k+1})^\wedge. \tag{4.37c}$$

It can be observed that, for $\boldsymbol{u}_k = \boldsymbol{0}$, the forced discrete Euler-Lagrange (4.36) and Hamilton (4.37) equations reduce to the corresponding unforced equations (4.31), (4.33) derived previously.

4.3 Multi-body Systems

The previous sections considered the discrete mechanics of a single rigid body whose configuration evolves directly on a Lie group. For multibody systems, the individual rigid bodies are connected through mechanical joints. These joints impose holonomic constraints on the configuration and therefore have to be included in the variational principle. In this work, constrained multibody systems are represented in maximal coordinates, following the formulation used for variational integrators in maximal coordinates in [BM21a] and for constrained Lie group variational dynamics in [TJVG25].

In minimal coordinates, the configuration is described directly by the independent degrees of freedom of the mechanism. This representation is compact, but the resulting kinematics and dynamics may contain nonlinear trigonometric expressions and dense coupling terms. This is disadvantageous for the polynomial optimization formulation considered later in this thesis. In contrast, maximal coordinates assign each rigid body its own position and orientation variables. The reduction to the physical degrees of freedom is then achieved by imposing algebraic joint constraints with Lagrange multipliers. Therefore, the constraints are not artificial state constraints, but hard mechanical constraints induced by the joints of the system [BM21b].

For a multi-body system with n_b rigid bodies in dimension $d \in \{2, 3\}$, the maximal-coordinate configuration manifold is given by

$$Q_{\max} = \prod_{i=1}^{n_b} \left(\mathbb{R}^d \times SO(d)\right). \tag{4.38}$$

Each rigid body i is described by its center of mass position $\boldsymbol{x}_i \in \mathbb{R}^d$ and its orientation $\boldsymbol{R}_i \in SO(d)$. Consequently, the full configuration is written as

$$q = (\boldsymbol{x}_1, \boldsymbol{R}_1, \ldots, \boldsymbol{x}_{n_b}, \boldsymbol{R}_{n_b}) \in Q_{\max}. \tag{4.39}$$

The joint structure is encoded by holonomic constraints

$$\boldsymbol{\phi}(q) = \mathbf{0}, \qquad \boldsymbol{\phi} : Q_{\max} \to \mathbb{R}^{n_c}, \tag{4.40}$$

where n_c denotes the number of scalar constraint equations.

A typical holonomic joint constraint can be written by equating two points attached to different rigid bodies. This is illustrated in Fig. 4.1. Let $\boldsymbol{\rho}_i$ and $\boldsymbol{\rho}_j$ denote constant vectors from the corresponding center of mass to the joint location in each body-fixed frame. Then a link-connecting constraint is given by

$$\boldsymbol{\phi}_{ij}(q) = \boldsymbol{x}_i + \boldsymbol{R}_i\boldsymbol{\rho}_i - \boldsymbol{x}_j - \boldsymbol{R}_j\boldsymbol{\rho}_j = \mathbf{0}. \tag{4.41}$$

To include the holonomic constraints in the discrete variational principle, the discrete action sum is augmented by Lagrange multipliers $\boldsymbol{\lambda}_k \in \mathbb{R}^{n_c}$. Using a trapezoidal

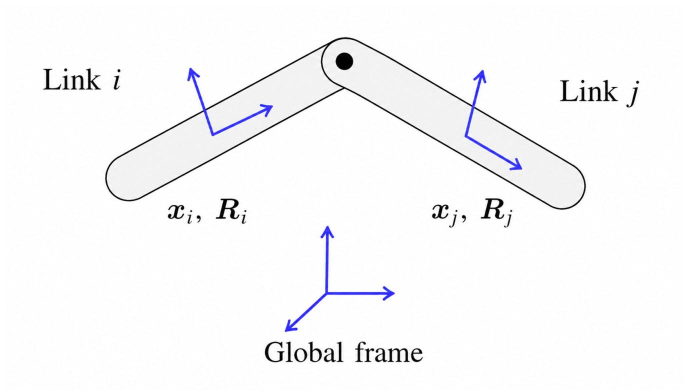


Figure 4.1: Two rigid bodies connected by a holonomic joint in maximal coordinates. Adapted from [BM21a].

approximation for the constraint contribution gives

$$S_d = \sum_{k=0}^{N-1} \left[L_d(q_k, q_{k+1}) + \frac{h}{2} \left(\boldsymbol{\lambda}_k^\top \boldsymbol{\phi}(q_k) + \boldsymbol{\lambda}_{k+1}^\top \boldsymbol{\phi}(q_{k+1}) \right) \right]. \tag{4.42}$$

The variation with respect to q_k, together with the discrete Lagrange–d'Alembert force contribution introduced in Section **??**, yields the forced and constrained discrete Euler-Lagrange equations

$$D_1 L_d(q_k, q_{k+1}) + D_2 L_d(q_{k-1}, q_k) + \boldsymbol{u}_k - h\, \boldsymbol{G}(q_k)^\top \boldsymbol{\lambda}_k = 0, \qquad \boldsymbol{\phi}(q_k) = \mathbf{0}, \tag{4.43}$$

where

$$\boldsymbol{G}(q_k) = \frac{\partial \boldsymbol{\phi}(q_k)}{\partial q_k} \tag{4.44}$$

denotes the constraint Jacobian. Equation (4.43) provides the general forced and constrained discrete Euler-Lagrange equation used for deriving multi-body dynamics.

4.3.1 Acrobot

Configuration Manifold The Acrobot model, illustrated in Fig. 4.2, consists of two planar rigid links connected by a revolute joint. The first link is attached to the inertial frame by a fixed revolute joint, while the second link is connected to the first link at the elbow. In contrast to the 3D-pendulum, the system is described in maximal coordinates. Therefore, each link is assigned its own center of mass position and orientation.

For each link $i \in \{1, 2\}$, the center of mass position is denoted by $\boldsymbol{x}_i \in \mathbb{R}^2$, and the orientation of the body-fixed frame with respect to the inertial frame is

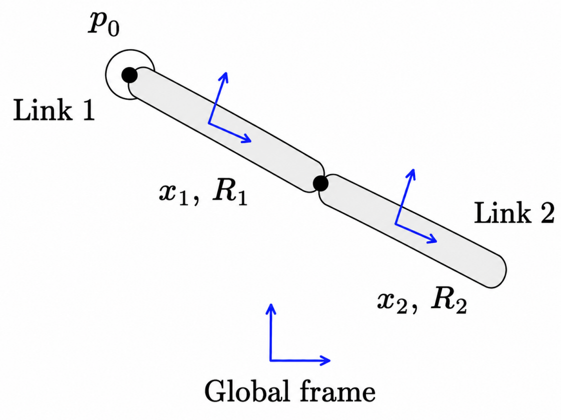


Figure 4.2: Acrobot configuration manifold in maximal coordinates.

described by the rotation matrix $\boldsymbol{R}_i \in SO(2)$. The special orthogonal group in two dimensions is defined as

$$SO(2) = \left\{ \boldsymbol{R} \in \mathbb{R}^{2\times 2} \mid \boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}_2,\ \det(\boldsymbol{R}) = 1 \right\}. \tag{4.45}$$

Every element $\boldsymbol{R}_i \in SO(2)$ can be written in terms of an angle $\theta_i \in \mathbb{R}$ as

$$\boldsymbol{R}_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{bmatrix}. \tag{4.46}$$

Thus, the unconstrained maximal-coordinate configuration manifold is given by

$$Q_{\max} = \left(\mathbb{R}^2 \times SO(2)\right)^2. \tag{4.47}$$

The configuration of the Acrobot is written as

$$q = (\boldsymbol{x}_1, \boldsymbol{R}_1, \boldsymbol{x}_2, \boldsymbol{R}_2) \in Q_{\max}. \tag{4.48}$$

Here, $\boldsymbol{R}_i$ maps vectors from the i-th body-fixed frame to the inertial frame.

Let $\boldsymbol{\rho}_{1,0} \in \mathbb{R}^2$ denote the vector from the center of mass of link 1 to the base joint, expressed in the first body-fixed frame. Furthermore, let $\boldsymbol{\rho}_{1,12} \in \mathbb{R}^2$ and $\boldsymbol{\rho}_{2,12} \in \mathbb{R}^2$ denote the vectors from the corresponding centers of mass to the elbow joint, expressed in the body-fixed frames of link 1 and link 2. The fixed base joint is imposed by

$$\boldsymbol{\phi}_0(q) = \boldsymbol{x}_1 + \boldsymbol{R}_1\boldsymbol{\rho}_{1,0} - \boldsymbol{p}_0 = \boldsymbol{0}, \tag{4.49}$$

where $\boldsymbol{p}_0 \in \mathbb{R}^2$ is the fixed base position in the inertial frame. The elbow joint is imposed by requiring the two joint points on link 1 and link 2 to coincide, yielding

$$\boldsymbol{\phi}_{12}(q) = \boldsymbol{x}_1 + \boldsymbol{R}_1\boldsymbol{\rho}_{1,12} - \boldsymbol{x}_2 - \boldsymbol{R}_2\boldsymbol{\rho}_{2,12} = \boldsymbol{0}. \tag{4.50}$$

Consequently, the full holonomic constraint vector is defined as

$$\boldsymbol{\phi}(q) = \begin{bmatrix} \boldsymbol{\phi}_0(q) \\ \boldsymbol{\phi}_{12}(q) \end{bmatrix} = \boldsymbol{0}. \tag{4.51}$$

In the discrete setting, the attitude of each link is updated by a relative rotation $\boldsymbol{F}_{i,k} \in SO(2)$, such that

$$\boldsymbol{R}_{i,k+1} = \boldsymbol{R}_{i,k}\boldsymbol{F}_{i,k}, \qquad i \in \{1,2\}. \tag{4.52}$$

The discrete configuration at time step k is therefore

$$q_k = (\boldsymbol{x}_{1,k}, \boldsymbol{R}_{1,k}, \boldsymbol{x}_{2,k}, \boldsymbol{R}_{2,k}) \in Q_c. \tag{4.53}$$

Continuous Lagrangian To derive the continuous Lagrangian of the Acrobot in maximal coordinates, the kinetic and potential energy of both rigid links are introduced. For each link $i \in \{1,2\}$, the translational velocity of the center of mass is denoted by $\dot{\boldsymbol{x}}_i = \boldsymbol{v}_i \in \mathbb{R}^2$. Let $\boldsymbol{M}_i = m_i\boldsymbol{I}_2$ be the mass matrix of link i, and let $\boldsymbol{J}_{d,i} \in \mathbb{R}^{2\times 2}$ denote the non-standard inertia matrix, similar to Section 4.2.1. Then, the kinetic energy with its translational T_T and rotational T_R components is written as

$$T = \underbrace{\sum_{i=1}^{2} \frac{1}{2}\boldsymbol{v}_i^\top \boldsymbol{M}_i \boldsymbol{v}_i}_{T_T} + \underbrace{\sum_{i=1}^{2} \frac{1}{2}\mathrm{tr}\left[\hat{\omega}_i \boldsymbol{J}_{d,i} \hat{\omega}_i^\top\right]}_{T_R}. \tag{4.54}$$

The gravitational potential energy is expressed directly by the center of mass positions. In contrast to the 3D-pendulum, zero potential is assumed for the rotational setting [BM21b], which gives

$$V_R(\boldsymbol{R}_1, \boldsymbol{R}_2) = 0. \tag{4.55}$$

Therefore, the gravitational potential energy is given by the translational case

$$V_T(\boldsymbol{x}_1, \boldsymbol{x}_2) = \sum_{i=1}^{2} g\, \boldsymbol{e}_2^\top \boldsymbol{M}_i \boldsymbol{x}_i, \tag{4.56}$$

where $\boldsymbol{e}_2 \in \mathbb{R}^2$ denotes the vertical unit vector in the inertial frame. Consequently, the continuous Lagrangian $L : TQ_{\max} \to \mathbb{R}$ is defined as

$$L(q,\dot{q}) = \sum_{i=1}^{2} \left(\frac{1}{2}\boldsymbol{v}_i^\top \boldsymbol{M}_i \boldsymbol{v}_i + \frac{1}{2}\mathrm{tr}\left[\hat{\omega}_i \boldsymbol{J}_{d,i} \hat{\omega}_i^\top\right] \right) - \sum_{i=1}^{2} g\, \boldsymbol{e}_2^\top \boldsymbol{M}_i \boldsymbol{x}_i. \tag{4.57}$$

Discrete Lagrangian For the Acrobot in maximal coordinates, the translational and rotational parts are discretized separately, following the structure of maximal-coordinate variational integrators in [BM21b]. Therefore, the discrete Lagrangian is written as

$$L_d(q_k, q_{k+1}) = T_{d,T}(q_k, q_{k+1}) + T_{d,R}(q_k, q_{k+1}) - V_{d,T}(q_k, q_{k+1}). \tag{4.58}$$

For the translational part, the center of mass velocity of each link is approximated by

$$\boldsymbol{v}_{i,k} \approx \frac{\boldsymbol{x}_{i,k+1} - \boldsymbol{x}_{i,k}}{h}, \qquad i \in \{1, 2\}. \tag{4.59}$$

[BM21b] defines the discrete translational kinetic energy contribution to be

$$T_{d,T}(q_k, q_{k+1}) = \sum_{i=1}^{2} \frac{1}{2h} \left(\boldsymbol{x}_{i,k+1} - \boldsymbol{x}_{i,k}\right)^\top \boldsymbol{M}_i \left(\boldsymbol{x}_{i,k+1} - \boldsymbol{x}_{i,k}\right). \tag{4.60}$$

Substituting the angular velocity approximation (4.23) into the rotational kinetic energy T_R of (4.54), the discrete rotational kinetic energy contribution is derived to

$$\begin{aligned} T_{d,R}(q_k, q_{k+1}) &= \sum_{i=1}^{2} \frac{1}{2h} \mathrm{tr}\left[\left(\boldsymbol{F}_{i,k} - \boldsymbol{I}_2\right) \boldsymbol{J}_{d,i} \left(\boldsymbol{F}_{i,k} - \boldsymbol{I}_2\right)^\top\right] \\ &= \sum_{i=1}^{2} \frac{1}{h} \mathrm{tr}\left[\left(\boldsymbol{I}_2 - \boldsymbol{F}_{i,k}\right) \boldsymbol{J}_{d,i}\right]. \end{aligned} \tag{4.61}$$

which calculation corresponds to the discrete rotational kinetic energy of the 3D-pendulum in Section 4.2.1.

Finally, using the trapezoidal rule for the discretization of the potential energy integral, the discrete translational potential contribution $V_{d,T}$ is given by

$$V_{d,T}(q_k, q_{k+1}) = \frac{h}{2} \sum_{i=1}^{2} m_i g \boldsymbol{e}_2^\top \left(\boldsymbol{x}_{i,k} + \boldsymbol{x}_{i,k+1}\right). \tag{4.62}$$

Combining the translational (4.60), rotational (4.61), and potential (4.62) contributions gives the discrete Lagrangian

$$\begin{aligned} L_d(q_k, q_{k+1}) &= L_d\left(\boldsymbol{x}_{1,k}, \boldsymbol{x}_{1,k+1}, \boldsymbol{x}_{2,k}, \boldsymbol{x}_{2,k+1}, \boldsymbol{F}_{1,k}, \boldsymbol{F}_{2,k}\right) \\ &= \sum_{i=1}^{2} \frac{1}{2h} \left(\boldsymbol{x}_{i,k+1} - \boldsymbol{x}_{i,k}\right)^\top \boldsymbol{M}_i \left(\boldsymbol{x}_{i,k+1} - \boldsymbol{x}_{i,k}\right) \\ &\quad + \sum_{i=1}^{2} \frac{1}{h} \mathrm{tr}\left[\left(\boldsymbol{I}_2 - \boldsymbol{F}_{i,k}\right) \boldsymbol{J}_{d,i}\right] \\ &\quad - \frac{h}{2} \sum_{i=1}^{2} g \boldsymbol{e}_2^\top \boldsymbol{M}_i \left(\boldsymbol{x}_{i,k} + \boldsymbol{x}_{i,k+1}\right). \end{aligned} \tag{4.63}$$

Discrete Euler-Lagrange Equations The discrete equations of motion are obtained from the stationary condition of the constrained discrete action. The holonomic constraints introduced in (4.51) are included by extending the action integral with Lagrange multipliers. Let

$$\boldsymbol{\lambda} = \begin{bmatrix} \boldsymbol{\lambda}_0 \\ \boldsymbol{\lambda}_{12} \end{bmatrix} \in \mathbb{R}^4, \tag{4.64}$$

where $\boldsymbol{\lambda}_0 \in \mathbb{R}^2$ enforces the base constraint and $\boldsymbol{\lambda}_{12} \in \mathbb{R}^2$ enforces the elbow constraint. Using the constrained action sum defined in (4.42), the variation is written as

$$\delta S_d = \delta \sum_{k=0}^{N-1} \left[L_d(q_k, q_{k+1}) + \frac{h}{2} \left(\boldsymbol{\lambda}_k^\top \boldsymbol{\phi}(q_k) + \boldsymbol{\lambda}_{k+1}^\top \boldsymbol{\phi}(q_{k+1}) \right) \right] = 0. \tag{4.65}$$

The translational and rotational components are treated separately, following the maximal-coordinate variational-integrator structure. In [BM21b], the variation of the translational component is given as

$$\delta_{\boldsymbol{x}_{i,k+1}} S_{d,c} = \left\langle \frac{1}{h} \boldsymbol{M}_i \left(\boldsymbol{v}_{i,k} - \boldsymbol{v}_{i,k+1} \right) - g \boldsymbol{M}_i \boldsymbol{e}_2 + \boldsymbol{G}_{x_i,k+1}^\top \boldsymbol{\lambda}_{k+1}, \delta \boldsymbol{x}_{i,k+1} \right\rangle. \tag{4.66}$$

using the relation (4.59). The constraint contribution is calculated to

$$\boldsymbol{G}_{x,k+1}^\top \boldsymbol{\lambda}_{k+1} = \begin{bmatrix} \dfrac{\partial \boldsymbol{\phi}_{0,k+1}}{\partial \boldsymbol{x}_{1,k+1}} & \dfrac{\partial \boldsymbol{\phi}_{0,k+1}}{\partial \boldsymbol{x}_{2,k+1}} \\ \dfrac{\partial \boldsymbol{\phi}_{12,k+1}}{\partial \boldsymbol{x}_{1,k+1}} & \dfrac{\partial \boldsymbol{\phi}_{12,k+1}}{\partial \boldsymbol{x}_{2,k+1}} \end{bmatrix}^\top \begin{bmatrix} \boldsymbol{\lambda}_{0,k+1} \\ \boldsymbol{\lambda}_{12,k+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\lambda}_{0,k+1} + \boldsymbol{\lambda}_{12,k+1} \\ -\boldsymbol{\lambda}_{12,k+1} \end{bmatrix}. \tag{4.67}$$

with the discretized constraints

$$\boldsymbol{\phi}(q_{k+1}) = \begin{bmatrix} \boldsymbol{\phi}_{0,k+1} \\ \boldsymbol{\phi}_{12,k+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{x}_{1,k+1} + \boldsymbol{R}_{1,k+1} \boldsymbol{\rho}_{1,0} - \boldsymbol{p}_0 \\ \boldsymbol{x}_{1,k+1} + \boldsymbol{R}_{1,k+1} \boldsymbol{\rho}_{1,12} - \boldsymbol{x}_{2,k+1} - \boldsymbol{R}_{2,k+1} \boldsymbol{\rho}_{2,12} \end{bmatrix}. \tag{4.68}$$

Therefore, substituting the gradient (4.67) into (4.66), the translational discrete Euler-Lagrange equations for link 1 and link 2 are

$$\frac{1}{h} \boldsymbol{M}_1 \left(\boldsymbol{v}_{1,k+1} - \boldsymbol{v}_{1,k} \right) + g \boldsymbol{M}_1 \boldsymbol{e}_2 - \left(\boldsymbol{\lambda}_{0,k+1} + \boldsymbol{\lambda}_{12,k+1} \right) = \boldsymbol{0}, \tag{4.69a}$$

$$\frac{1}{h} \boldsymbol{M}_2 \left(\boldsymbol{v}_{2,k+1} - \boldsymbol{v}_{2,k} \right) + g \boldsymbol{M}_2 \boldsymbol{e}_2 + \boldsymbol{\lambda}_{12,k+1} = \boldsymbol{0}. \tag{4.69b}$$

For the rotational part, the derivatives of **??** have to be taken for each link, analogously to the 3D-pendulum of Section 4.2.1. Firstly, the directional derivative $\left\langle D_{\boldsymbol{F}_{i,k}} L_d(\boldsymbol{F}_{i,k}), \delta \boldsymbol{f}_{i,k} \right\rangle$ is given by applying (4.3), which yields to

$$\begin{aligned} \left\langle T_e^* L_{\boldsymbol{F}_{i,k}} D_{\boldsymbol{F}_{i,k}} L_{d,R,i}, \hat{\chi}_k \right\rangle &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_d \left(\boldsymbol{F}_{i,k} \exp(\epsilon \hat{\chi}_k) \right) \\ &= -\frac{1}{h} \operatorname{tr} \left[\boldsymbol{F}_{i,k} \hat{\chi}_k \boldsymbol{J}_{d,i} \right] \\ &= \left\langle \frac{1}{h} \left(\boldsymbol{J}_{d,i} \boldsymbol{F}_{i,k} - \boldsymbol{F}_{i,k}^\top \boldsymbol{J}_{d,i} \right), \hat{\chi}_k \right\rangle \\ &= \left\langle \boldsymbol{\Pi}_{i,k}, \hat{\chi}_k \right\rangle, \end{aligned} \tag{4.70}$$

corresponding to the derivative of the single body 3D-pendulum derived in Section 4.2.1. Here,

$$\boldsymbol{\Pi}_{i,k} = \frac{1}{h} \left(\boldsymbol{J}_{d,i} \boldsymbol{F}_{i,k} - \boldsymbol{F}_{i,k}^\top \boldsymbol{J}_{d,i} \right), \qquad i \in \{1, 2\}, \tag{4.71}$$

denotes the discrete angular momentum of link i associated with the relative rotation$\boldsymbol{F}_{i,k}$. It is the analogue of the discrete Legendre momentum (4.32a) used for the 3D-pendulum in Section 4.2.1, now defined separately for each rigid link [Lee08].

Since $SO(2)$ is abelian, it follows that $\mathrm{Ad}_{\boldsymbol{F}_{i,k}} = I$ [Lee08]. Therefore, the coadjoint term is equal to (4.70) at time step $k+1$, which gives

$$\mathrm{Ad}^*_{\boldsymbol{F}^{-1}_{i,k+1}} \left(T^*_e L_{\boldsymbol{F}_{i,k+1}} D_{\boldsymbol{F}_{i,k+1}} L_d(\boldsymbol{F}_{i,k+1})\right) = \frac{1}{h} \left(\boldsymbol{J}_{d,i} \boldsymbol{F}_{i,k+1} - \boldsymbol{F}^\top_{i,k+1} \boldsymbol{J}_{d,i}\right). \tag{4.72}$$

It remains to derive the directional derivative of the rotations $R_{i,k+1}$. From (4.65), it is observed that only the constraints $\boldsymbol{\phi}(q_k)$ are dependent on the rotation matrices $\boldsymbol{R}_{1,k}$ and $\boldsymbol{R}_{2,k}$. Hence, the variation of the constraint contribution

$$\left\langle D_{\boldsymbol{R}_{i,k+1}}(h\boldsymbol{\lambda}^\top_{k+1}\boldsymbol{\phi}_{k+1}), \delta\boldsymbol{R}_{i,k+1}\right\rangle = \left\langle T^*_e L_{\boldsymbol{R}_{i,k+1}} D_{\boldsymbol{R}_{i,k+1}}(h\boldsymbol{\lambda}^\top_{k+1}\boldsymbol{\phi}_{k+1}), \hat{\eta}_{i,k+1}\right\rangle,$$

has to be evaluated for each link i, using (4.1). For link $i = 1$, the variation of the constraint contribution is given by

$$\begin{aligned}
&\left\langle T^*_e L_{\boldsymbol{R}_{1,k+1}} D_{\boldsymbol{R}_{1,k+1}} \left(h\boldsymbol{\lambda}^\top_{k+1}\boldsymbol{\phi}_{k+1}\right), \hat{\eta}_{1,k+1}\right\rangle \\
&= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} h\left[\boldsymbol{\lambda}^\top_{0,k+1}\boldsymbol{\phi}^\epsilon_{0,k+1} + \boldsymbol{\lambda}^\top_{12,k+1}\boldsymbol{\phi}^\epsilon_{12,k+1}\right] \\
&= h\left[\boldsymbol{\lambda}^\top_{0,k+1}\boldsymbol{R}_{1,k+1}\hat{\eta}_{1,k+1}\boldsymbol{\rho}_{1,0} + \boldsymbol{\lambda}^\top_{12,k+1}\boldsymbol{R}_{1,k+1}\hat{\eta}_{1,k+1}\boldsymbol{\rho}_{1,12}\right] \\
&= h\left[\left(\boldsymbol{\rho}_{1,0} \times \boldsymbol{R}^\top_{1,k+1}\boldsymbol{\lambda}_{0,k+1}\right) + \left(\boldsymbol{\rho}_{1,12} \times \boldsymbol{R}^\top_{1,k+1}\boldsymbol{\lambda}_{12,k+1}\right)\right] \cdot \eta_{1,k+1} \\
&= \left\langle h\left[\left(\boldsymbol{\rho}_{1,0} \times \boldsymbol{R}^\top_{1,k+1}\boldsymbol{\lambda}_{0,k+1}\right) + \left(\boldsymbol{\rho}_{1,12} \times \boldsymbol{R}^\top_{1,k+1}\boldsymbol{\lambda}_{12,k+1}\right)\right]^\wedge, \hat{\eta}_{1,k+1}\right\rangle,
\end{aligned} \tag{4.73}$$

which calculation correspponts to the derivative of the 3D-pendulum with respect to the rotation matrix R_k derived in Section 4.2.1. In contrast, the variation of the constraint contribution for link $i = 2$ only depends on $\phi_{12,k+1}$ and is derived to

$$\begin{aligned}
&\left\langle T^*_e L_{\boldsymbol{R}_{2,k+1}} D_{\boldsymbol{R}_{2,k+1}} \left(h\boldsymbol{\lambda}^\top_{k+1}\boldsymbol{\phi}_{k+1}\right), \hat{\eta}_{2,k+1}\right\rangle \\
&= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} h\left[\boldsymbol{\lambda}^\top_{12,k+1}\boldsymbol{\phi}^\epsilon_{12,k+1}\right] \\
&= \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} h\left[\boldsymbol{\lambda}^\top_{12,k+1}\left(-\boldsymbol{R}_{2,k+1}\exp(\epsilon\hat{\eta}_{2,k+1})\boldsymbol{\rho}_{2,12}\right)\right] \\
&= -h\left[\boldsymbol{\lambda}^\top_{12,k+1}\boldsymbol{R}_{2,k+1}\hat{\eta}_{2,k+1}\boldsymbol{\rho}_{2,12}\right] \\
&= -h\left(\boldsymbol{\rho}_{2,12} \times \boldsymbol{R}^\top_{2,k+1}\boldsymbol{\lambda}_{12,k+1}\right) \cdot \eta_{2,k+1} \\
&= \left\langle -h\left(\boldsymbol{\rho}_{2,12} \times \boldsymbol{R}^\top_{2,k+1}\boldsymbol{\lambda}_{12,k+1}\right)^\wedge, \hat{\eta}_{2,k+1}\right\rangle.
\end{aligned} \tag{4.74}$$

The full rotational contribution is described, by substituting (4.70), (4.72), (4.73) and (4.74) for each link $i = 1, 2$ into (**??**). Moreover, (**??**) gives the translational description. Finally, the full discrete Euler-Lagrange equations of the unforced Acrobot in maximal coordinates are expressed on $\left(\mathbb{R}^2 \times SO(2)\right)^2$ as

Translation:

$$\frac{1}{h}\boldsymbol{M}_1\left(\boldsymbol{v}_{1,k+1}-\boldsymbol{v}_{1,k}\right)+g\boldsymbol{M}_1\boldsymbol{e}_2-\left(\boldsymbol{\lambda}_{0,k+1}+\boldsymbol{\lambda}_{12,k+1}\right)=\mathbf{0}, \tag{4.75a}$$

$$\frac{1}{h}\boldsymbol{M}_2\left(\boldsymbol{v}_{2,k+1}-\boldsymbol{v}_{2,k}\right)+g\boldsymbol{M}_2\boldsymbol{e}_2+\boldsymbol{\lambda}_{12,k+1}=\mathbf{0}, \tag{4.75b}$$

Rotation:

$$\boldsymbol{\Pi}_{1,k}-\boldsymbol{\Pi}_{1,k+1}+h\left[\left(\boldsymbol{\rho}_{1,0}\times\boldsymbol{R}_{1,k+1}^{\top}\boldsymbol{\lambda}_{0,k+1}\right)+\left(\boldsymbol{\rho}_{1,12}\times\boldsymbol{R}_{1,k+1}^{\top}\boldsymbol{\lambda}_{12,k+1}\right)\right]^{\wedge}=\mathbf{0}, \tag{4.75c}$$

$$\boldsymbol{\Pi}_{2,k}-\boldsymbol{\Pi}_{2,k+1}-h\left[\left(\boldsymbol{\rho}_{2,12}\times\boldsymbol{R}_{2,k+1}^{\top}\boldsymbol{\lambda}_{12,k+1}\right)\right]^{\wedge}=\mathbf{0}, \tag{4.75d}$$

Constraints:

$$\boldsymbol{\phi}_{0,k+1}=\boldsymbol{x}_{1,k+1}+\boldsymbol{R}_{1,k+1}\boldsymbol{\rho}_{1,0}-\boldsymbol{p}_0=\mathbf{0}, \tag{4.75e}$$

$$\boldsymbol{\phi}_{12,k+1}=\boldsymbol{x}_{1,k+1}+\boldsymbol{R}_{1,k+1}\boldsymbol{\rho}_{1,12}-\boldsymbol{x}_{2,k+1}-\boldsymbol{R}_{2,k+1}\boldsymbol{\rho}_{2,12}=\mathbf{0}, \tag{4.75f}$$

Kinematics:

$$\boldsymbol{R}_{1,k+1}=\boldsymbol{R}_{1,k}\boldsymbol{F}_{1,k}, \tag{4.75g}$$

$$\boldsymbol{R}_{2,k+1}=\boldsymbol{R}_{2,k}\boldsymbol{F}_{2,k}. \tag{4.75h}$$

in dependence to its constraints (4.51).

Controlled Acrobot As in the controlled 3D-pendulum case (Section 4.2.1), the trapezoidal approximation distributes the virtual work equally to both endpoints of the interval. Therefore, the discrete Legendre force contributions of (4.10) are given by

$$u_{d,k}^{-}=u_{d,k}^{+}=\frac{h}{2}u_k. \tag{4.76}$$

The Acrobot is actuated by a single motor located at the elbow joint [BM21a]. In contrast to the controlled 3D-pendulum, the input is therefore not an external moment acting on one rigid body, but an internal joint torque acting between the two links. The generalized torque vector is defined as

$$\boldsymbol{\tau}_k=\begin{bmatrix}-u_k\\ u_k\end{bmatrix}, \tag{4.77}$$

with $u_k \in \mathbb{R}$ being the scalar elbow torque, applying opposite moments between the two links.

Since the Acrobot is actuated by a motor at the elbow joint, the control input is a pure rotational torque acting between the two links. Hence, there is no external translational force acting directly on the centers of mass, and the translational force contribution is set to

$$\boldsymbol{f}_{1,k}^{x}=\boldsymbol{f}_{2,k}^{x}=\mathbf{0}. \tag{4.78}$$

Using the discrete Lagrange-d'Alembert principle (4.12a), the generalized torque contribution (4.77) is added only to the unforced rotational equations (4.75c) and (4.75d). Therefore, the controlled rotational discrete Euler-Lagrange equations are derived to

$$\mathbf{\Pi}_{1,k} - \mathbf{\Pi}_{1,k+1} + h\left[\left(\boldsymbol{\rho}_{1,0} \times \boldsymbol{R}_{1,k+1}^{\top}\boldsymbol{\lambda}_{0,k+1}\right) + \left(\boldsymbol{\rho}_{1,12} \times \boldsymbol{R}_{1,k+1}^{\top}\boldsymbol{\lambda}_{12,k+1}\right) - u_{k+1}\right]^{\wedge} = \mathbf{0}, \tag{4.79a}$$

$$\mathbf{\Pi}_{2,k} - \mathbf{\Pi}_{2,k+1} + h\left[u_{k+1} - \left(\boldsymbol{\rho}_{2,12} \times \boldsymbol{R}_{2,k+1}^{\top}\boldsymbol{\lambda}_{12,k+1}\right)\right]^{\wedge} = \mathbf{0}. \tag{4.79b}$$

For $u_{k+1} = 0$, these equations reduce to the corresponding unforced rotational equations derived previously.

Chapter 5

Certifiable Trajectory Optimization via Semidefinite Relaxation

5.1 Trajectory Optimization with Variational Integrators

5.1.1 Optimization Problem Formulation

5.2 Semidefinite Optimization and Relaxation

5.2.1 Semidefinite Relaxation and the Moment-SOS Hierarchy

Sum of Squares (SOS) Relaxation

Moment Relaxation

5.2.2 Sparse Relaxations

5.3 Optimization Formulation

5.3.1 Control of 3D Pendulum

$$
\begin{aligned}
\min_{\substack{\{R_k\}_{k=0}^{N} \\ \{F_k\}_{k=0}^{N-1}, \{u_{p,k}\}_{k=1}^{N-1}}} \quad & \rho_R \left\| R_N - R_N^{\mathrm{des}} \right\|_F^2 + \rho_F \left\| F_{N-1} - F^{\mathrm{des}} \right\|_F^2 \\
& + \sum_{k=0}^{N-1} \alpha_R \left\| R_k - R_N^{\mathrm{des}} \right\|_F^2 + \sum_{k=0}^{N-2} \alpha_F \left\| F_k - F^{\mathrm{des}} \right\|_F^2 \\
& + \sum_{k=1}^{N-1} \gamma^{-1} \left\| u_{p,k} \right\|_2^2 . && (5.1a) \\
\text{s.t.} \quad & \delta_2 f_{32,k+1} - \delta_3 f_{23,k+1} - \delta_3 f_{32,k} + \delta_2 f_{23,k} - h^2 (M_{1,k+1} + u_{1,k+1}) = 0, && (5.1b) \\
& \delta_3 f_{13,k+1} - \delta_1 f_{31,k+1} - \delta_1 f_{13,k} + \delta_3 f_{31,k} - h^2 (M_{2,k+1} + u_{2,k+1}) = 0, && (5.1c) \\
& \delta_1 f_{21,k+1} - \delta_2 f_{12,k+1} - \delta_2 f_{21,k} + \delta_1 f_{12,k} - h^2 (M_{3,k+1} + u_{3,k+1}) = 0, && (5.1d) \\
& (R_{k+1})_{ij} - \sum_{\ell=1}^{3} (R_k)_{i\ell} (F_k)_{\ell j} = 0, && (5.1e) \\
& \|r_{1,k}\|_2^2 - 1 = \|r_{2,k}\|_2^2 - 1 = \|r_{3,k}\|_2^2 - 1 = 0, && (5.1f) \\
& r_{1,k}^\top r_{2,k} = r_{2,k}^\top r_{3,k} = r_{1,k}^\top r_{3,k} = 0, && (5.1g) \\
& r_{1,k} \times r_{2,k} - r_{3,k} = r_{2,k} \times r_{3,k} - r_{1,k} = r_{3,k} \times r_{1,k} - r_{2,k} = 0_{3\times 1}, && (5.1h) \\
& \|f_{1,k}\|_2^2 - 1 = \|f_{2,k}\|_2^2 - 1 = \|f_{3,k}\|_2^2 - 1 = 0, && (5.1i) \\
& f_{1,k}^\top f_{2,k} = f_{2,k}^\top f_{3,k} = f_{1,k}^\top f_{3,k} = 0, && (5.1j) \\
& f_{1,k} \times f_{2,k} - f_{3,k} = f_{2,k} \times f_{3,k} - f_{1,k} = f_{3,k} \times f_{1,k} - f_{2,k} = 0_{3\times 1}, && (5.1k) \\
& \frac{\mathrm{tr}(F_k) - 1}{2} \geq f_{c,\min}, && (5.1l) \\
& u_{\max}^2 - \|u_{p,k}\|_2^2 \geq 0. && (5.1m)
\end{aligned}
$$

$$
\text{with} \qquad M_{k+1} = \begin{bmatrix} M_{1,k+1} \\ M_{2,k+1} \\ M_{3,k+1} \end{bmatrix} = mg \begin{bmatrix} \rho_2 r_{33,k+1} - \rho_3 r_{32,k+1} \\ \rho_3 r_{31,k+1} - \rho_1 r_{33,k+1} \\ \rho_1 r_{32,k+1} - \rho_2 r_{31,k+1} \end{bmatrix}, \tag{5.2}
$$

$$
u_{k+1} = \begin{bmatrix} u_{1,k+1} \\ u_{2,k+1} \\ u_{3,k+1} \end{bmatrix} = \begin{bmatrix} r_{32,k+1} u_{p3,k+1} - r_{33,k+1} u_{p2,k+1} \\ r_{33,k+1} u_{p1,k+1} - r_{31,k+1} u_{p3,k+1} \\ r_{31,k+1} u_{p2,k+1} - r_{32,k+1} u_{p1,k+1} \end{bmatrix}, \tag{5.3}
$$

$$R_k = [r_{1,k}\ r_{2,k}\ r_{3,k}] \in \mathbb{R}^{3\times 3}, \qquad r_{1,k}, r_{2,k}, r_{3,k} \in \mathbb{R}^{3\times 1}, \tag{5.4}$$

$$F_k = [f_{1,k}\ f_{2,k}\ f_{3,k}] \in \mathbb{R}^{3\times 3}, \qquad f_{1,k}, f_{2,k}, f_{3,k} \in \mathbb{R}^{3\times 1}, \tag{5.5}$$

$$J_d = \mathrm{diag}(\delta_1, \delta_2, \delta_3), \qquad \rho_c = \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix}, \qquad u_{p,k} = \begin{bmatrix} u_{p1,k} \\ u_{p2,k} \\ u_{p3,k} \end{bmatrix}. \tag{5.6}$$

In (5.1b)–(5.1d), the discrete Euler–Lagrange constraints hold for $k = 0, \ldots, N-2$. The kinematic constraint (5.1e) holds for $i, j = 1, 2, 3$ and $k = 0, \ldots, N-1$. The constraints (5.1f)–(5.1h) hold for $k = 0, \ldots, N$, while (5.1i)–(5.1k), (5.1l), and (5.1m) hold for $k = 0, \ldots, N-1$.

Chapter 6

Evaluation

6.1 Numerical Simulation Results

In this section, numerical simulations are conducted to evaluate the correctness of the derived discrete integrators for mechanical models on Lie groups. In particular, the structure-preserving properties of the variational discretizations derived in Section **??** are examined. The analysis focuses especially on the preservation of the momentum map associated with the symmetry of the system, the evolution on the Lie-group configuration manifold, and the long-time behaviour of the total energy. This validation is important since discrete mechanical models obtained by variational integration provide a reliable basis for certifiable trajectory optimization. If the discrete flow violates the manifold structure or introduces artificial drift in conserved quantities such as momentum, the resulting optimization model becomes less faithful to the underlying mechanics [Lee08]. In contrast, variational integrators are constructed from a discrete variational principle and are symplectic, momentum preserving, and consist robust long-time energy behaviour [MW01]. For the evaluation, the evolution of the total energy, the momentum-map behaviour, and the preservation of the Lie-group structure of the 3D-pendulum are compared with the classical fourth-order Runge-Kutta method and a projected Runge-Kutta on $SO(3)$ as reference integrators [CO03, Hai00] (see Table. 6.1).

Table 6.1: Comparison of the considered integration methods properties.

Method	**Symplectic**	**Manifold Preserving**
Variational Integrator	✓	✓
RK4	✗	✗
Projected RK4	✗	✓

Runge–Kutta 4th order As a basline integrator, the classical explicit fourth-order Runge-Kutta (RK4) is applied to the continuous rigid-body equations written as as first-order system $(\dot{\boldsymbol{R}}, \dot{\boldsymbol{\omega}}) = \boldsymbol{f}(\boldsymbol{R}, \boldsymbol{\omega})$. For a system of the form

$$\dot{\boldsymbol{y}} = \boldsymbol{f}(\boldsymbol{y}), \tag{6.1}$$

the classical RK4 scheme computes the four stage values

$$(\boldsymbol{k}_1^R, \boldsymbol{k}_1^\omega) = \boldsymbol{f}(\boldsymbol{R}_n, \boldsymbol{\omega}_n), \tag{6.2a}$$

$$(\boldsymbol{k}_2^R, \boldsymbol{k}_2^\omega) = \boldsymbol{f}\left(\boldsymbol{R}_n + \frac{h}{2}\boldsymbol{k}_1^R, \boldsymbol{\omega}_n + \frac{h}{2}\boldsymbol{k}_1^\omega\right), \tag{6.2b}$$

$$(\boldsymbol{k}_3^R, \boldsymbol{k}_3^\omega) = \boldsymbol{f}\left(\boldsymbol{R}_n + \frac{h}{2}\boldsymbol{k}_2^R, \boldsymbol{\omega}_n + \frac{h}{2}\boldsymbol{k}_2^\omega\right), \tag{6.2c}$$

$$(\boldsymbol{k}_4^R, \boldsymbol{k}_4^\omega) = \boldsymbol{f}\left(\boldsymbol{R}_n + h\boldsymbol{k}_3^R, \boldsymbol{\omega}_n + h\boldsymbol{k}_3^\omega\right), \tag{6.2d}$$

and updates the numerical solution according to

$$\boldsymbol{R}_{n+1} = \boldsymbol{R}_n + \frac{h}{6}\left(\boldsymbol{k}_1^R + 2\boldsymbol{k}_2^R + 2\boldsymbol{k}_3^R + \boldsymbol{k}_4^R\right), \tag{6.3a}$$

$$\boldsymbol{\omega}_{n+1} = \boldsymbol{\omega}_n + \frac{h}{6}\left(\boldsymbol{k}_1^\omega + 2\boldsymbol{k}_2^\omega + 2\boldsymbol{k}_3^\omega + \boldsymbol{k}_4^\omega\right), \tag{6.3b}$$

for each discrete-time step n [But16].

The explicit RK4 method does not preserve the orthogonality constraint $\boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{I}$ and leaves the manifold configuration of $SO(3)$, since the rotation matrix $\boldsymbol{R}_n$ is updated additively in the vector space of $\mathbb{R}^{3\times3}$ [CO03].

RK4 projected to $SO(3)$ To satisfy the orthogonality constraint

$$SO(3) = \left\{\boldsymbol{R} \in \mathbb{R}^{3\times3} \;\middle|\; \boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{I}_3,\ \det(\boldsymbol{R}) = 1\right\}.$$

the explicit RK4 method is combined with a projection of the discrete-time updated rotation matrix $\boldsymbol{R}_n$ on $SO(3)$. Therefore, after performing the RK4 step in $\mathbb{R}^{3\times3}$ the obtained rotation matrix $\widetilde{\boldsymbol{R}}_{n+1}$, the singular value decomposition is used

$$\widetilde{\boldsymbol{R}}_{n+1} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top, \tag{6.4}$$

following a reconstruction to derive the updated rotation matrix

$$\boldsymbol{R}_{n+1} = \boldsymbol{U}\boldsymbol{V}^\top, \tag{6.5}$$

which is projected on $SO(3)$ [YTT11]. Consequently, $\boldsymbol{R}_{n+1}$ satisfy the orthogonality constraint, while the angular velocity $\boldsymbol{\omega}_{n+1}$ remains unchanged on the vector space $\mathbb{R}^{3\times3}$. However, this method does not inherit structure preserving properties of the Lie group variational integrator, for example, symplecticity or momentum conservation [CO03].

6.1.1 3D Pendulum

For evaluating the geometric properties preservation of the previously defined 3D-pendulum (see Section **??**), numerical simulations are conducted for the unforced and controlled cases. The rigid body is modeled as a uniform solid cylinder of length $l = 1\,\mathrm{m}$ and radius $r = 5\,\mathrm{cm}$, with its center of mass at $\frac{l}{2}$. Thereby, the physical parameters characterizing the configuration manifold, are chosen as

$$m = 1\,\mathrm{kg}, \qquad g = 9.81\,\mathrm{m\,s^{-2}}, \tag{6.6}$$

$$\boldsymbol{\rho}_c = \begin{bmatrix} 0 & 0 & 0.5 \end{bmatrix}^\top \mathrm{m}, \qquad \boldsymbol{e}_3 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^\top. \tag{6.7}$$

The inertia matrix about the center of mass can be taken from the inertia table given in [Ric]. Here, the rotation of the center of mass in x and y direction are represented about the central diameter $\boldsymbol{J}_{11,com} = \boldsymbol{J}_{22,com} = \frac{1}{12}m(3r^2 + 4l^2)$, and the rotation of the center of mass in gravity direction z is perpendicular to the axis of rotation $\boldsymbol{J}_{33,com} = \frac{1}{2}mr^2$. The off-diagonal terms are equal to zero, because the body-fixed frame is aligned with the principal axes of the cylindrically symmetric body [Lee08]. Additionally, the inertia matrix about the pivot in body-frame is obtained by the parallel axis theorem of [BR18] given by

$$\boldsymbol{J} = \boldsymbol{J}_{\mathrm{com}} + m\left(\|\boldsymbol{\rho}_c\|^2\boldsymbol{I} - \boldsymbol{\rho}_c\boldsymbol{\rho}_c^\top\right). \tag{6.8}$$

After substituting the chosen values, the 3D-pendulum's inertia matrix is given as

$$\boldsymbol{J} = \begin{bmatrix} 0.33396 & 0 & 0 \\ 0 & 0.33396 & 0 \\ 0 & 0 & 0.00125 \end{bmatrix} \mathrm{kg\,m^2}. \tag{6.9}$$

Given (3.7), the non-standard inertia matrix is calculated to

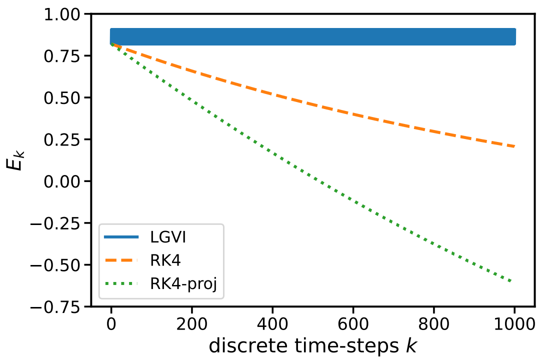
$$\boldsymbol{J}_d = \begin{bmatrix} 0.000625 & 0 & 0 \\ 0 & 0.000625 & 0 \\ 0 & 0 & 0.333333 \end{bmatrix} \mathrm{kg\,m^2}. \tag{6.10}$$

Using the introduced computational approach of Section 4.1.6, the derived discrete-time implicit Hamilton equations of the unforced (4.33) and controlled (4.37) 3D-pendulum are converted to an equivalent vector equation in $\mathbb{R}^3$ using the Cayley transformation. From [CJ12], the corresponding nonlinear vector equation $0 = \boldsymbol{A}(\boldsymbol{f})$ of the 3D-pendulum is taken as

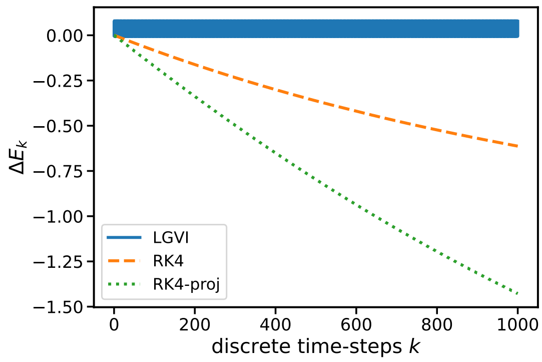
$$0 = \boldsymbol{a} + \boldsymbol{a} \times \boldsymbol{f} + \boldsymbol{f}(\boldsymbol{a}^\top\boldsymbol{f}) - 2\boldsymbol{J}\boldsymbol{f} = \boldsymbol{A}(f), \tag{6.11}$$

with $\boldsymbol{a} = h(\boldsymbol{\Pi}_k + \frac{h}{2}\boldsymbol{M}_k)$. Additionally, [CJ12] defines its Jacobian $\nabla A(\boldsymbol{f})$ to

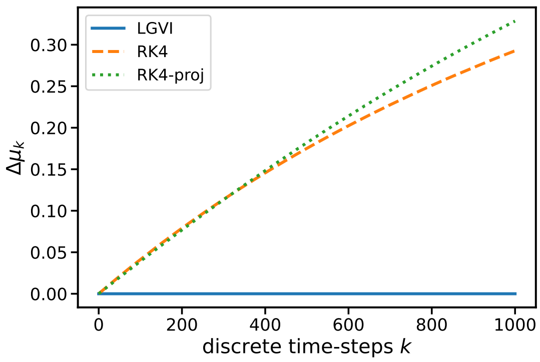
$$\nabla A(\boldsymbol{f}) = \hat{\boldsymbol{a}} + (\boldsymbol{a}^\top\boldsymbol{f})\boldsymbol{I} + \boldsymbol{f}\boldsymbol{a}^\top - 2\boldsymbol{J}. \tag{6.12}$$



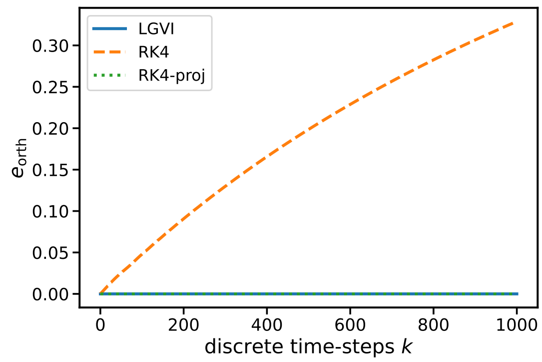
(a) Total energy E_k.



(b) Energy deviation ΔE_k.



(c) Momentum error $\Delta\mu_k$.



(d) Orthogonality error e_{orth}.

Figure 6.1: Comparison of the numerical preservation properties for the unforced 3D pendulum.

The tolerance for the Newton iteration is defined to $\left\|f^{(i+1)} - f^{(i)}\right\| < 10^{-12}$. The initial conditions are taken from [Lee08]

$$\boldsymbol{f}^0 = (2\boldsymbol{J} - \hat{\boldsymbol{a}})^{-1}\boldsymbol{a}, \qquad \boldsymbol{R}_0 = \boldsymbol{I}, \qquad \boldsymbol{\Omega}_0 = \begin{bmatrix} 4.14 & 4.14 & 4.14 \end{bmatrix}^\top \mathrm{rad\,s^{-1}}, \tag{6.13}$$

where $\boldsymbol{f}^0$ is the linearized solution of (6.11). The corresponding implementation is documented in the GitHub repository [Els26].

Unforced 3D-Pendulum

The results of the numerical simulation for the unforced case, with $\boldsymbol{u}_k = \boldsymbol{0}$, are shown in Fig. 6.1. The simulation is performed over 1000 discrete-time steps with a step size of $h = 0.05\,\mathrm{s}$. Thereby, the accuracy of the different integrator methods, given in Table 6.1, are compared to each other, by plotting the corresponding total energy

$$E_k = \frac{1}{2}\boldsymbol{\Omega}_k^\top \boldsymbol{J}\boldsymbol{\Omega}_k - mg\,\boldsymbol{e}_3^\top \boldsymbol{R}_k \boldsymbol{\rho}_c,$$

and its deviation $\Delta E_k = E_k - E_0$, the angular momentum error along the gravity direction

$$\Delta\mu_k = \boldsymbol{e}_3^\top \boldsymbol{R}_k \boldsymbol{\Pi}_k - \boldsymbol{e}_3^\top \boldsymbol{R}_0 \boldsymbol{\Pi}_0,$$

defined in [Lee08], and the orthogonality error

$$e_{\text{orth},k} = \left\| \boldsymbol{I} - \boldsymbol{R}_k^\top \boldsymbol{R}_k \right\|_F .$$

The total energy (see Fig. 6.1 (a)) and the corresponding energy deviation (see Fig. 6.1 (b)) illustrates the preserving energy behavior over a long-time in discrete-time flow for the Lie Group Variational Integrator (LGVI). It remains close to the reference level with only small bounded oscillations, because of their symplectic strcuture, which are the exact solution of a pertubed Hamiltonian [Hai94]. However, the RK4 and projected RK4 display a clear drift of the energy away from its real value, which remains constant for this model, as the system is unforced.

Moreover, the momentum error (see Fig. 6.1 (c)) confirms that the LGVI method preserves the momentum, validated for continuous time in 3.2.2. The computational results of RK4 and its projection on $SO(3)$ show that these methods are not reliable drifting from the momentum initial value, as they not preserve the total linear momentum [Lee08]. Especially, the projected RK4 consists of a stronger discrepancy, as it fixes $\boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{I}$, while the angular velocity $\boldsymbol{\omega}$ remains unchanged, which influences the momentum.

Lastly, the orthogonality error highlights that the structure of the Lie group configuration manifold is well preserved in the LGVI, which is depictured in Fig. 6.1 (d). Similar result is obtained from the projected RK4, while the classic explicit RK4 violates the orthogonality constraint. For RK4 e_{orth} increases linearly with respect to the simulation time-steps.

Consequently, only the LGVI preserves the manifold structure together with the momentum and energy behaviour over a long-time horizon. While RK4 can be projected to the Lie group manifold of $SO(3)$, it can not preserve the momentum and consists of drifting energy behaviour over long-time.

Controlled 3D-Pendulum

In the controlled case, the previously derived free dynamics of the 3D-pendulum are extended by an external body-fixed control moment. Following defined control structure of (4.34), the time-dependent control parameter is chosen as

$$\boldsymbol{u}_p(t) = \begin{bmatrix} 0.15\sin(0.5t) \\ 0.10\cos(0.25t) \\ 0 \end{bmatrix} . \tag{6.14}$$

The corresponding discrete control input is obtained by evaluating the control law at the discrete time nodes $\boldsymbol{u}_{p,k} = \boldsymbol{u}_p(t_k)$. Thus, the control input acts as a body-fixed external moment that is orthogonal to the gravity direction expressed in the body frame, given by $\boldsymbol{R}_k^\top \boldsymbol{e}_3$. Consequently, the control moment has no component along this direction.

For the RK4 simulation, the continuous-time Hamilton equations (3.16) are used, adding the contol input $\boldsymbol{u(t)}$, based on Lagrange-d' Alembert principle of Section 4.1.5. With the body angular momentum defined by $\boldsymbol{\Pi} = \boldsymbol{J}\boldsymbol{\Omega}$, the forced continuous-time equations of motion are written as

$$\dot{\boldsymbol{\Pi}} = mg\,\boldsymbol{\rho}_c \times \boldsymbol{R}^\top \boldsymbol{e}_3 - \boldsymbol{J}^{-1}\boldsymbol{\Pi} \times \boldsymbol{\Pi} + \boldsymbol{u}(t), \tag{6.15a}$$

$$\dot{\boldsymbol{R}} = \boldsymbol{R}\left(\boldsymbol{J}^{-1}\boldsymbol{\Pi}\right)^\wedge. \tag{6.15b}$$

In contrast to the unforced case, the external input acts on the system. Therefore, the total energy and the momentum map are no longer expected to be conserved.

For the Lie group variational integrator, the forced discrete Hamilton equations derived in (4.37) are applied. The implicit equation for the relative update $\boldsymbol{F}_k$ is again solved by using the Cayley transformation. Compared to the unforced case, only the vector $\boldsymbol{a}$ in (6.11) changes due to the additional control input. For the controlled dynamics, it is given by

$$\boldsymbol{a} = h\left(\boldsymbol{\Pi}_k + \frac{h}{2}\left(\boldsymbol{M}_k + \boldsymbol{u}_k\right)\right). \tag{6.16}$$

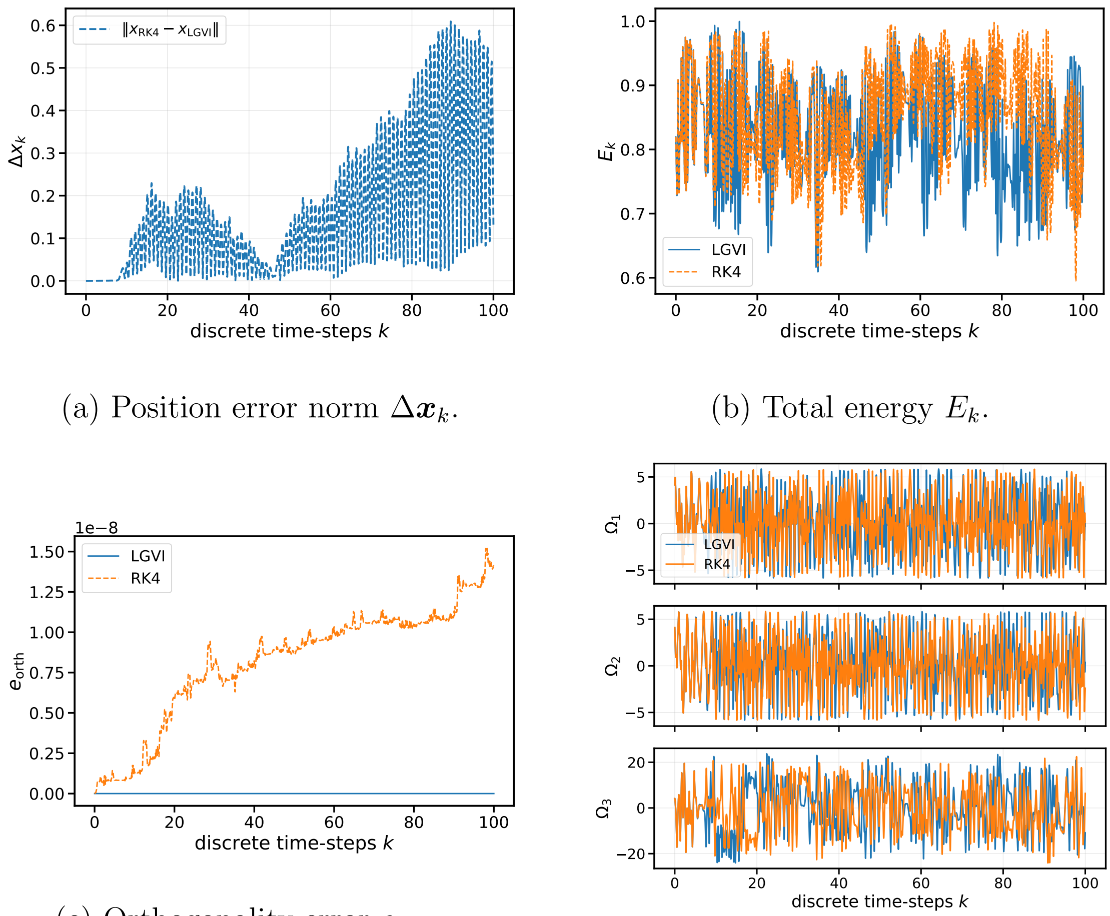
Therefore, the nonlinear vector equation (6.11) and its Jacobian (6.12) are solved equaly for the unforced simulation, with the modified vector $\boldsymbol{a}$ including the control contribution. After computing $\boldsymbol{f}_k$, the relative attitude update is obtained from the Cayley transformation and the discrete-time states are propagated according to the forced LGVI. The corresponding implementation is documented in the GitHub repository [Els26]

The numerical simulation of the controlled 3D-pendulum compares the classical RK4 method with the forced LGVI. Since the purpose of this experiment is to evaluate the behaviour of the forced variational discretization under an external input, the projected RK4 method is not considered in this comparison. The same physical parameters and initial conditions as in the unforced case are used.

Fig. 6.2 shows the numerical results for the controlled 3D-pendulum. The simulation is performed over $N = 100$ discrete-time steps with step size $h = 0.001\,\mathrm{s}$. This shorter time interval is chosen because the differences between the RK4 method and the forced LGVI already become visible after a small number of time steps. During approximately the first eight time steps, both methods show almost identical behaviour, which indicates that the derived forced equations and their implementation are consistent. As the simulation progresses, however, the trajectories begin to separate.

The position error in Fig. 6.2 (a) is computed from the center-of-mass position $\boldsymbol{x}_k = \boldsymbol{R}_k\boldsymbol{\rho}_c$, as

$$\Delta\boldsymbol{x}_k = \left\|\boldsymbol{x}_{k,\mathrm{RK4}} - \boldsymbol{x}_{k,\mathrm{LGVI}}\right\|_2. \tag{6.17}$$



(a) Position error norm $\Delta \boldsymbol{x}_k$.

(b) Total energy E_k.

(c) Orthogonality error $e_{\mathrm{orth},k}$.

(d) Angular velocity components $\boldsymbol{\Omega}_k$.

Figure 6.2: Comparison of the forced LGVI and RK4 simulation for the controlled 3D pendulum.

The plot shows that the position error $\Delta \boldsymbol{x}_k$ is initially close to zero, but increases over the simulation. Therefore, both methods reproduce the same short-time motion, while their trajectories separate after a certain number of time steps. This behaviour is expected for forced rigid-body dynamics, where small numerical differences in the attitude $\boldsymbol{R}_k$ and momentum update $\boldsymbol{\Pi}_k$ can accumulate over time.

The total energy is shown in Fig. 6.2 (b). In contrast to the unforced case, the total mechanical energy is not expected to be conserved, since the external input $u_{p,k}$ acts on the system. It is observed that energetic behaviour of LGVI and RK4 under the same applied control input follow a similar trend at the beginning, but small deviations appear as the trajectories begin to separate, which is consistent with the position error.

The orthogonality error $e_{\mathrm{orth},k}$ is illustrated in Fig. 6.2 (c). Again, the plot confirms that the LGVI preserves the Lie group structure staying on the manifold by updating the attitude through $\boldsymbol{R}_{k+1} = \boldsymbol{R}_k\boldsymbol{F}_k$ with $\boldsymbol{F}_k \in SO(3)$. For RK4, the attitude is updated additively in the surrounding vector space using the discrete update map (6.3). Over the short interval and due to the small step size h, the orthogonality error remains small, but it is not preserved and increased over time, especially for a long time horizon and smaller step sizes h.

Finally, Fig. 6.2 (d) compares the angular velocity components $\boldsymbol{\Omega}_k$. In the discrete Hamiltonian formulation, the angular velocity is recovered from the body angular momentum by

$$\hat{\boldsymbol{\Omega}}_k = (\boldsymbol{J}^{-1}\boldsymbol{\Pi}_k)^{\wedge}, \tag{6.18}$$

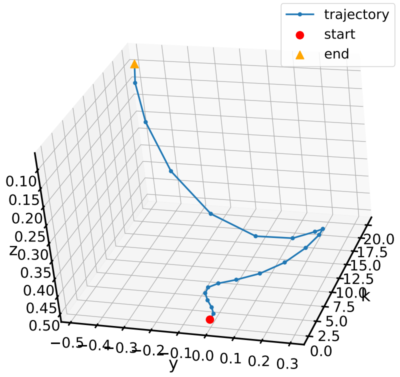
discretizing and solving (3.15) for $\hat{\boldsymbol{\Omega}}_k$. The angular velocity components of LGVI and RK4 are equal at the beginning of the simulation, but begin to differ after a small number of steps. This confirms the same behaviour observed in the position error $\Delta\boldsymbol{x}_k$ and the ploted Energy E_k. The 3D-Pendulum numerical simulation using external control inputs produces consistent short-time dynamics of LGVI and RK4, while the numerical trajectories between both methods separate as the simulation continues.

6.2 Optimization Results

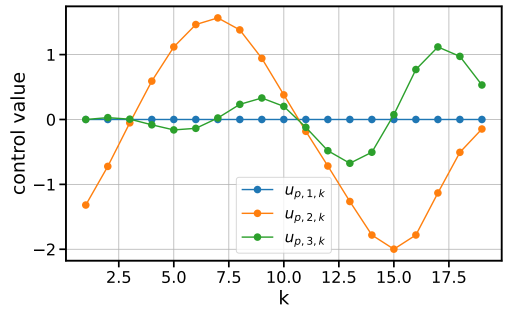
SPOT-Based Implementation

6.2.1 Controlled 3D-Pendulum

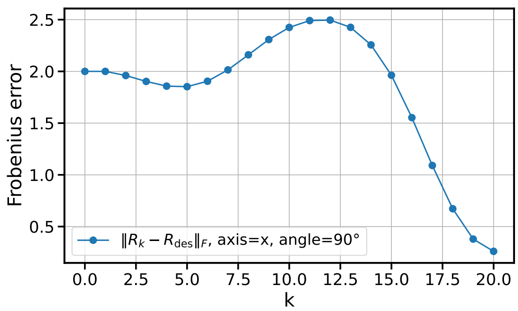
Minimal degree chordal extension



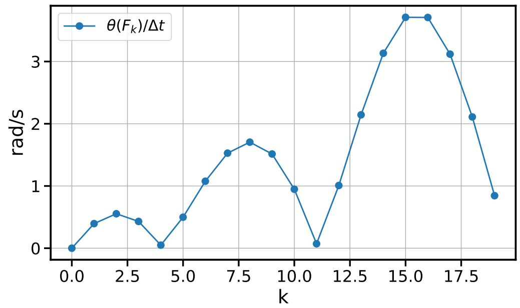
(a) Center-of-mass trajectory.



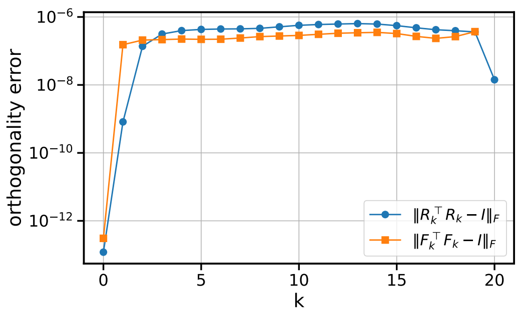
(b) Control inputs $\boldsymbol{u}_{p,k}$.



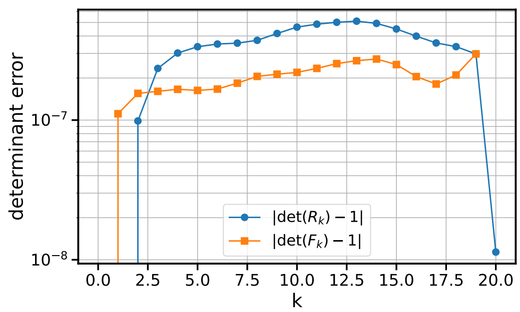
(c) Frobenius tracking error $\|\boldsymbol{R}_k - \boldsymbol{R}_{\mathrm{des}}\|_F$.



(d) Relative rotation speed $\theta(\boldsymbol{F}_k)/h$.



(e) Orthogonality errors of $\boldsymbol{R}_k$ and $\boldsymbol{F}_k$.



(f) Determinant errors of $\boldsymbol{R}_k$ and $\boldsymbol{F}_k$.

Figure 6.3: SPOT optimization results for the controlled 3D pendulum performing a 90° rotation about the x-axis using the MD sparsity setting.

Chapter 7

Discussion

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Acronyms

BIBO	bounded-input bounded-output
HRC	human-robot collaboration
MPC	model-predictive control

Notation

$\mathbf{x}_n$	n-dimensional vector named x
$\mathbf{X}_{m\times n}$	$m \times n$ dimensional matrix named X

List of Symbols

$\mathbf{u}$	control input vector
$\mathbf{u}_k$	control input vector with time step
$\mathbf{x}_k$	state vector with time step

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